

Contents

5. Review of 1225	2
5.2. The Definite Integral	2
5.2.1. Motivation	2
5.5. The Substitution Rule	10
6. Applications of Integrals	15
6.1. Area Between Curves	15
6.2. Volumes by Cross-Sections, Disks, and Washers	20
6.2.1. Slicing by Parallel Planes	20
6.2.2. Solids of Revolution: The Disk method	27
6.2.3. Solids of Revolution: The Washer Method	29
6.3. Volumes by Cylindrical Shells	35
6.4. Work	41
6.4.1. Units	41
6.4.2. Non-Constant Force	42
6.4.3. Lifting Objects	42
6.4.4. Stretching and Compressing Springs	43
6.4.5. Pumping Liquids	46
6.5. Average Value of a Function	50
7. Integration Techniques	54
7.1. Integration by Parts	54
7.1.1. How to we choose which function to use for u ?	55
7.1.2. Indefinite Integrals	56
7.1.3. Definite Integrals	58
7.2. Trigonometric Integrals	62
7.3. Trigonometric Substitutions	69
7.3.1. Motivation	69
7.3.2. Substitutions To Use	72
7.3.3. Indefinite Integrals	73
7.3.4. Definite Integrals	79

5. Review of 1225

5.2. The Definite Integral

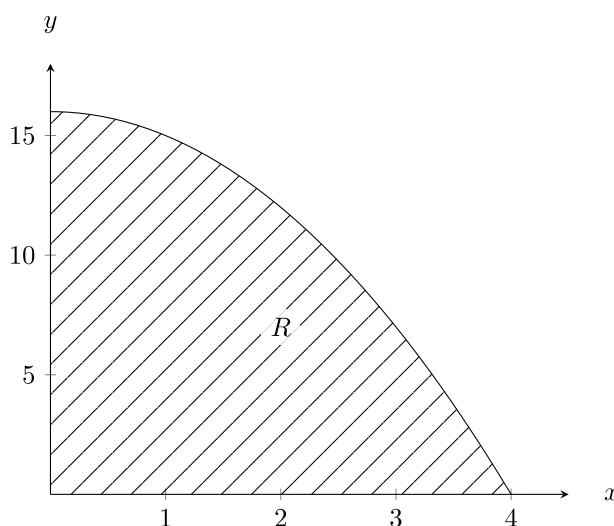
In this section, we will primarily review Riemann sums. This is because Riemann sums are used to derive many of the techniques used in the upcoming sections.

5.2 Learning Objectives:

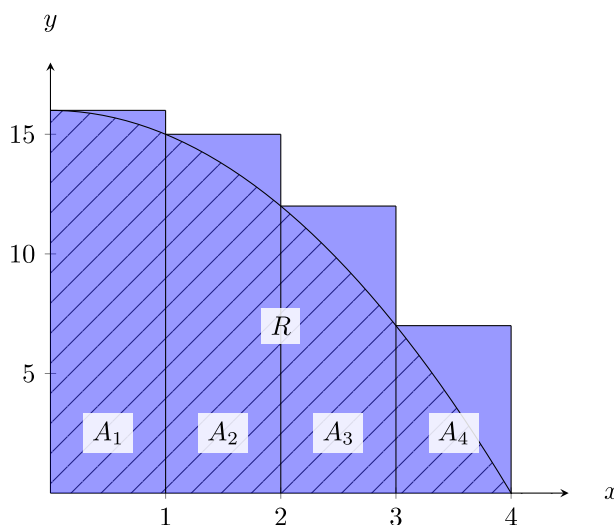
- I can approximate the (signed/unsigned) area under a curve using different types of Reimann sums.
- I can use more rectangles/sub-intervals to get a more accurate approximation.
- I know the definition of a definite integral in terms of the limit or Reimann sums.
- I can rewrite a limit of Reimann sums as a definite integral, and back.

5.2.1. Motivation

Suppose we wanted to calculate the area of a shaded region R that lies above the x -axis, below the graph of $f(x) = -x^2 + 16$, and between the vertical lines $x = 0$ and $x = 4$.



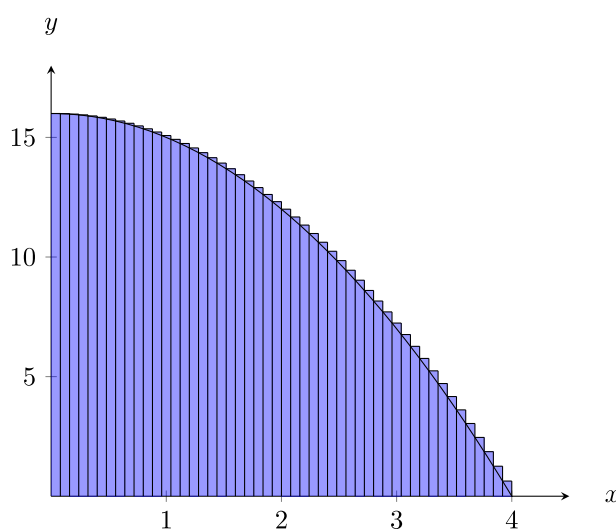
How might we estimate this area with rectangles?



The area can be estimated by summing the areas of the rectangles $R \approx A_1 + A_2 + A_3 + A_4$.

Remark

- We also aligned the left side of the rectangles to the curve, but we could have used any point between the left and right sides of the rectangles to get the height of the rectangles.
 - For example, right end point and midpoint are both popular choices.
- We used rectangles instead of another shape, because the area of a rectangle is easy to find. We only need width and height, and the height is just a single function evaluation.
 - For example, the area of A_1 is $A_1 = (1)(f(0)) = (1)(16) = 16$.
 - Another shape you may encounter (outside this class) is the trapezoid, using the left AND right end points of the domain to get the 2 required heights for a trapezoid.
- We used 4 rectangles, but any number of rectangles can be used.
 - Using more (smaller) rectangles improves the accuracy of our estimated area. See figure below.



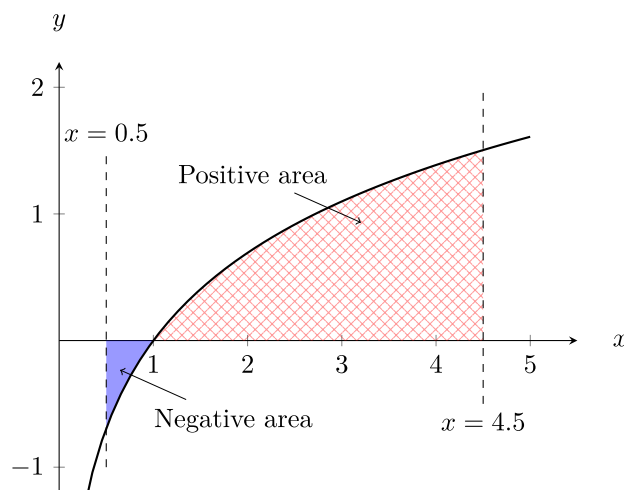
- In the above cases, the approximate areas have been overestimates of the actual area. In general, it is not possible to tell when we have an over or underestimate by looking at a picture. However, there are some cases we can always tell.
 - If the function is decreasing, then the right Riemann sums will underestimate and the left Riemann sums will overestimate.
 - If the function is increasing, then the right Riemann sums will overestimate and the left Riemann sums will underestimate.
 - If the function changes between increasing and decreasing, then we cannot guarantee one way without more information.

Exercise 5.2.1

1. Draw a few pictures of different increasing and decreasing functions and some left/right Riemann sums to get a feel for why they are always over/underestimates.
2. Does concavity have anything to do with the approximations being over or underestimates of the area under the curve?

Example 5.2.1

Suppose you want to find the *signed* area of the region R that lies between the x -axis, the graph of $f(x) = \ln(x)$, and between the vertical lines $x = 0.5$ and $x = 4.5$. The exact signed area would look something like this.

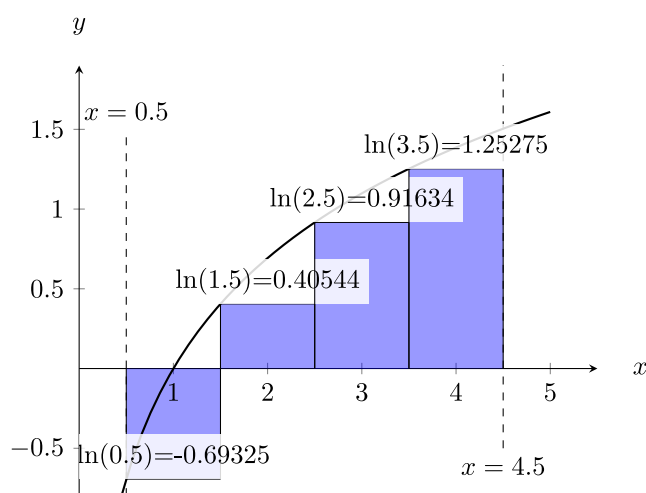


Approximate the signed area (as described above) to two decimal places with 4 rectangles using:

1. Left-hand Riemann sum
2. Right-hand Riemann sum
3. Mid-point Riemann sum

Solution

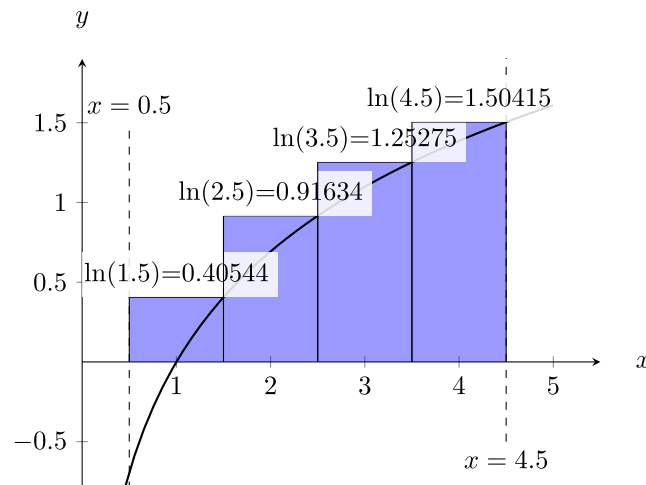
1. The left hand Reimann sum looks like the below image. Note, the area of the first rectangle is negative, because it is below the x -axis.



Thus,

$$\begin{aligned}
 L_4 &= A_1 + A_2 + A_3 + A_4 \\
 &= (1)(\ln(0.5)) + (1)(\ln(1.5)) + (1)(\ln(2.5)) + (1)(\ln(3.5)) \\
 &= \underbrace{1}_{\Delta x} (\ln(0.5) + \ln(1.5) + \ln(2.5) + \ln(3.5)) \\
 &\approx 1.88
 \end{aligned}$$

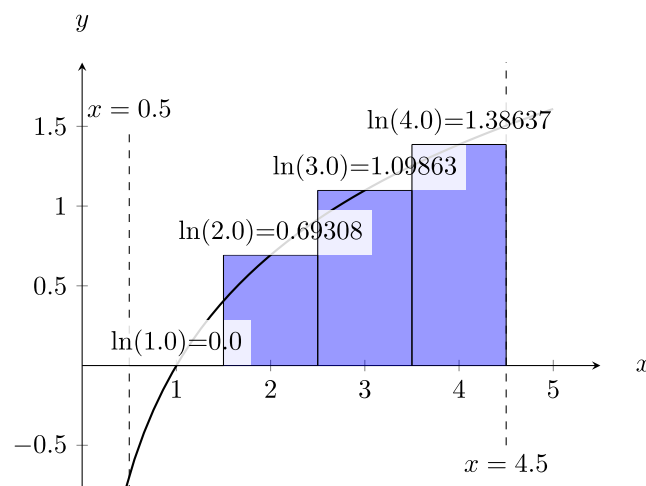
2. The right hand Reimann sum looks like the below image. Note, the area of the first rectangle is **no longer negative**, because it is now above the x -axis.



Thus,

$$\begin{aligned}
 R_4 &= (1)(\ln(1.5)) + (1)(\ln(2.5)) + (1)(\ln(3.5)) + (1)(\ln(4.5)) \\
 &= \underbrace{1}_{\Delta x} (\ln(1.5) + \ln(2.5) + \ln(3.5) + \ln(4.5)) \\
 &\approx 4.08
 \end{aligned}$$

3. The mid-point Reimann sum looks like the below image. Note, the first rectangle has height 0 since $\ln(1) = 0$.



Thus,

$$\begin{aligned}
 M_4 &= (1)(\ln(1)) + (1)(\ln(2)) + (1)(\ln(3)) + (1)(\ln(4)) \\
 &\approx 3.18
 \end{aligned}$$

Remark

- The actual signed area is 3.11 rounded to 2 decimal places.
- This does not give the (unsigned) area (ignoring the fact that some parts are negative).
- We can find the unsigned area (counting all area as positive) by using $-f(x)$ for the rectangle height for the points where $f(x)$ is negative.
- As before, using more rectangles can give us a more accurate estimate of the areas (signed and unsigned). *Let's push this idea to the limit...*

Definition 5.2.1

Let f be a function defined on the interval $[a, b]$. We can divide the interval $[a, b]$ into n sub-intervals of equal width $\Delta x = \frac{b-a}{n}$. Let $x_0 = a, x_1, x_2, \dots, x_{n-1}, x_n = b$ be the endpoints of these sub-intervals. Let $x_1^*, x_2^*, \dots, x_n^*$ be any sample points in these sub-intervals such that x_i^* is in the sub-interval $[x_{i-1}, x_i]$. Then the **definite integral of f from a to b** is defined as

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

Remark

- The definite integral $\int_a^b f(x) dx$ represents the **exact signed area** under the curve $y = f(x)$ between a and b .
- The integral symbol is a stretched “S” shape with the **lower** and **upper bound** above and below the symbol respectively (“S” is for sum). The dx at the end closes the integral but also indicates that we are integrating with respect to x .
 - Eventually we will have other variables or multiple variables we are integrating over, so this is important for clarity.
 - The variable we are integrating over says where the “base” of the rectangles will be.
- As we saw in the previous examples, as we use more rectangles to approximate the area, we get closer to the exact signed area we are looking for. In the definition of the definite integral, n is the number of rectangles, and when we take the limit as n goes to infinity we are subdividing into more and more rectangles to approximate the area. Eventually the rectangles approach “infinitesimally thin rectangles,” that add to the exact area we are looking for.
- We have been talking about left-hand, right-hand, and mid-point Riemann sums, but we could also use a point 10% of the way into the sub-interval, or 30% into the sub-interval, or anywhere inside each sub-interval. We write this using x_i^* to indicate a generic choice of left/right/mid-point sample point in the i -th rectangle. The i here is the most important part, since it means this point depends on which rectangle we are in. The star is also important, since without it the definition would imply it denotes the end-points of the sub-intervals.
 - You could theoretically use a different point for each rectangle, but we won't. Keep it simple and consistent. Use all left end points or all right end points etc depending on your choice for a problem.
- You should become comfortable translating between a definite integral and a limit of Riemann sums and back again. An example is below, come to office hours to discuss how starting at a Riemann sum does not lead to 1 correct definite integral, but many possible ones.

Example 5.2.2

Write

$$\int_1^3 \sqrt[3]{x^2 + 2} \, dx$$

as a limit of Riemann sums.

Solution

We are not told to use a left or right Riemann sum specifically, so we can choose either. We will do a right-hand Riemann sum here (Practice on your own: the left one). The **bounds** on the integral give us $a = 1$ and $b = 3$. We can also clearly see the function is

$$f(x) = \sqrt[3]{x^2 + 2}.$$

The next thing to do is to write the width of the rectangles in terms of n :

$$\Delta x = \frac{b - a}{n} = \frac{2}{n}.$$

Lastly, we need a way to represent the right-end points of the rectangles in terms of i and n , and we will try to find a pattern for this. The **left** side of the first rectangle is at 1, and the next is Δx to the right of it, the next is another Δx to the right, and so on. The **right** sides of the rectangles skip 1 and start at $1 + \Delta x$, then add another Δx every step. Repeated addition is multiplication so

$$x_i^* = 1 + \frac{2i}{n}.$$

All that's left is to put it altogether

$$\int_1^3 \sqrt[3]{x^2 + 2} \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt[3]{\left(1 + \frac{2i}{n}\right)^2 + 2} \left(\frac{2}{n}\right).$$

Example 5.2.3

Interpret

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2\left(3 + \frac{3i}{n}\right)}{1 + \left(3 + \frac{3i}{n}\right)^2} \left(\frac{3}{n}\right)$$

as a definite integral.

Solution

For Δx we are looking for a number over n to show up, sometimes multiplied by i . We see a $\frac{3}{n}$ in a few places that fits this bill, so we will say $\Delta x = \frac{3}{n}$. Then, we try to identify x_i^* , the sample points. These have $\Delta x i$ in them, we can pick just the $\frac{3i}{n}$ or the $3 + \frac{3i}{n}$. Either way we can get a correct answer. Let's pick $x_i^* = 3 + \frac{3i}{n}$, then **in this case** we must have

$$f(x_i^*) = \frac{2x_i^*}{1 + (x_i^*)^2}$$

(by replacing the $3 + \frac{3i}{n}$ with x_i^*). Lastly, all the sample points x_i^* “start” at 3 plus something: so $a = 3$. Then since $\Delta x = \frac{3}{n}$ we know the width of the interval is 3 (see numerator), so $b = 6$. Putting that all together

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{2(3 + \frac{3i}{n})}{1 + (3 + \frac{3i}{n})^2} \left(\frac{3}{n}\right) = \int_3^6 \frac{2x}{1 + x^2} dx.$$

Exercise 5.2.2

Redo the above problem but using $\Delta x = \frac{3i}{n}$. Notice how this changes the function and the bounds of the integral. To check your answer plug in both integrals into an integral calculator (<https://www.integral-calculator.com/> or a TI 84 or more should do it). You know you are correct if you get the same number for both!

Property 5.2.1 Properties of the Definite Integral

Let f and g be continuous functions. Let c be a constant.

- $\int_b^a f(x) dx = - \int_a^b f(x) dx$
- $\int_a^a f(x) dx = 0$
- $\int_a^b c dx = c(b - a)$
- $\int_a^b f(x) \pm g(x) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$
- $\int_a^b cf(x) dx = c \int_a^b f(x) dx$
- $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$
- If $f(x) \geq 0$ for all $a \leq x \leq b$, then $\int_a^b f(x) dx \geq 0$
- If $f(x) \geq g(x)$ for all $a \leq x \leq b$, then $\int_a^b f(x) dx \geq \int_a^b g(x) dx$
- If $m \leq f(x) \leq M$ for all $a \leq x \leq b$, then $m(b - a) \leq \int_a^b f(x) dx \leq M(b - a)$

Exercise 5.2.3

1. Rewrite in your own words what each property above is saying/does. You may have to think about some of these a while, and you may discuss in office hours.
2. Justify/Prove the third property using a sketch and a few words. ($\int_a^b c \, dx = c(b - a)$)
3. Justify to yourself why the penultimate property is true. (If $f(x) \geq g(x)$ for all $a \leq x \leq b$, then $\int_a^b f(x) \, dx \geq \int_a^b g(x) \, dx$)
4. Using the above properties prove the last property. (If $m \leq f(x) \leq M$ for all $a \leq x \leq b$, then $m(b - a) \leq \int_a^b f(x) \, dx \leq M(b - a)$) This should be done in 2 steps since there are 2 inequalities here. *This is not trivial, so spend a bit of time thinking and working on this. The process of doing this yourself should deepen your understanding of what is going on here, if you get stuck, bring your work to office hours to discuss.*

5.2 Section Summary:

- We discussed area beneath a curve, and discussed Reimann sums as ways to approximate the area.
- We discussed the difference between signed and unsigned area.
- We took the limit of Reimann sums to get the exact *signed* area, this became our definition of the definite integral.
- We practiced translating between Reimann sums and definite integrals.
- We looked at several properties of integrals.

5.5. The Substitution Rule

Recall that we linked the definite integral and area under a curve to the concept of the antiderivative. We did this using the fundamental theorem of Calculus.

5.5 Learning Objectives:

- I can use a substitution to undo the chain rule when antidifferentiating.
- I can change the bounds of a definite integral when I make a substitution.
- I can try a few different things for my substitution before giving up.

Theorem 5.5.1 The Fundamental Theorem of Calculus, Pt I

Let f be a function defined on the interval $[a, b]$. Let F be the function defined, for all x in $[a, b]$, by $F(x) = \int_a^x f(t) dt$. Then F is continuous on $[a, b]$ and differentiable on (a, b) and $F'(x) = f(x)$ for all x in (a, b) so F is an antiderivative of f .

Corollary 5.5.1 The Fundamental Theorem of Calculus, Pt II

If f is a continuous function defined on $[a, b]$ and F is an antiderivative of f in $[a, b]$ then

$$\int_a^b f(t) dt = F(b) - F(a).$$

This allows us to start evaluating definite integrals without worrying about limits of sums. Instead, to find the area under a curve, we just need an antiderivative of it.

Exercise 5.5.1

Find an antiderivative for the following functions. Check your work by taking the derivative of your answer.

1. $f(x) = x^3 - 2x + 1$
2. $g(x) = (x + 1)^{10}$

This is fine for simple functions that we can recognize that we are looking at the derivative of something. As functions get more complicated, we need more clever techniques for antideriving/integrating.

Theorem 5.5.2 The Substitution Rule (for Indefinite Integrals)

If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

$$\int f(g(x))g'(x) dx = \int f(u) du.$$

Note: The Substitution Rule is the “inverse” of the Chain Rule (for derivatives).

Remark

There are two main things we look for when deciding to use a substitution, although there may be other less common applications.

- First, we use a substitution when we see function composition, or a function plugged into another function.
- Another use case is if the derivative (up to scaling by a constant) of part of the integrand also appears in the integrand.

Tip

To integrate $\int f(g(x))g'(x) dx$:

1. Substitute $u = g(x)$ and $du = g'(x) dx$
2. Integrate with respect to u .
3. Replace u with $g(x)$ in the result.

Example 5.5.1

Find

$$\int \frac{2x+1}{x^2+x+3} dx$$

Solution

We notice that $\frac{d}{dx}(x^2+x+3) = 2x+1$. This is our indication to try a substitution:

$$\text{Let } u = x^2 + x + 3$$

$$\frac{du}{dx} = 2x + 1$$

$$du = 2x + 1 dx$$

Substituting in u and $2x+1 dx$ (note we could solve this for dx and plug that in instead) we get

$$\begin{aligned} \int \frac{2x+1}{x^2+x+3} dx &= \int \frac{1}{u} du \\ &= \ln|u| + C \\ &= \ln|x^2+x+3| + C \end{aligned}$$

So far we have only discussed indefinite integrals, those without bound. For definite integrals we have to be careful about the bounds when we make the substitution, since the bounds are in terms of the original variable, not our substituted variable.

Theorem 5.5.3 The Substitution Rule (for Definite Integrals)

If g' is continuous on $[a, b]$ and f is continuous on the range of $u = g(x)$, then

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Warning

When computing a definite integral with a substitution, you have two options: switch the bounds of integration to be with respect to u **or** switch the variable of integration back to x after finding the antiderivative and use the original bounds. Note, if you do not switch the bounds, you cannot write the old bounds in terms of x in your work on steps where you are talking about u .

Both methods will be demonstrated in this section (and probably throughout these notes), you are free to choose your favorite method, use either, etc.

Example 5.5.2

Evaluate $\int_1^e \frac{(\ln(x))^2}{2x} dx$.

Solution

We see a function composition, $\ln(x)$ looks like it has been plugged into something else. We pick that as u .

$$\text{Let } u = \ln(x)$$

$$du = \frac{1}{x} dx.$$

We also need to change the bounds of integration

$$u(e) = \ln(e) = 1$$

$$u(1) = \ln(1) = 0$$

The integral then is

$$\begin{aligned} \int_1^e \frac{(\ln(x))^2}{2x} dx &= \int_0^1 \frac{u^2}{2} du \\ &= \left[\frac{1}{6} u^3 \right]_0^1 \\ &= \frac{1}{6} \end{aligned}$$

Notice in the method above, since we changed the bounds of the integral to be in terms of u , we did not need to plug in $u = \ln(x)$ again. Changing the bounds tends to be faster in most examples, and if you don't change the bounds you have to be much more careful with notation.

Note

An optional bit of notation for clarity on bounds may be useful. To remember you have to change the bounds of integration when doing a substitution you can write the variable name in the lower bound. For example,

$$\int_{x=1}^e \frac{(\ln(x))^2}{2x} dx = \int_{u=0}^1 \frac{u^2}{2} du.$$

This kind of extra notation for clarity can act as training wheels as you get used to the substitution step of changing bounds, but also this notation becomes *very* useful in multivariable calculus.

Here are some less obvious substitution problems.

Example 5.5.3

Evaluate $\int_0^{\frac{\pi}{16}} \tan(4x) \, dx$.

Solution

It seems that $4x$ has been plugged into the tangent function. However, using this as our u would not change the fact that we don't have a known antiderivative for $\tan(u)$. To see the substitution, sometimes we need to rewrite problems

$$\int_0^{\frac{\pi}{16}} \tan(4x) \, dx = \int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)} \, dx$$

Now we can see a function and (a constant multiple of) its derivative: $\frac{d}{dx} \cos(4x) = -4 \sin(4x)$. This becomes our candidate for u

$$\begin{aligned} \text{Let } u &= \cos(4x) \\ \text{then } du &= -4 \sin(4x) \, dx \\ -\frac{1}{4} du &= \sin(4x) \, dx \end{aligned}$$

(notice we can solve for dx and plug that in, but after we sub u we can see we have the $\sin(4x) \, dx$ there anyway) and the bounds are changed to

$$\begin{aligned} u\left(\frac{\pi}{16}\right) &= \frac{\sqrt{2}}{2} \\ u(0) &= 1 \end{aligned}$$

then

$$\int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)} \, dx = \int_1^{\frac{\sqrt{2}}{2}} -\frac{1}{4} \frac{1}{u} \, du.$$

The antiderivative of $\frac{1}{u}$ is $\ln(u)$. So,

$$\begin{aligned} \int_0^{\frac{\pi}{16}} \frac{\sin(4x)}{\cos(4x)} \, dx &= \left[-\frac{1}{4} \ln|u| \right]_1^{\frac{\sqrt{2}}{2}} \\ &= -\frac{1}{4} \ln\left(\frac{\sqrt{2}}{2}\right) \end{aligned}$$

(Optionally,) This could be further simplified by $-\frac{1}{4} \ln\left(\frac{\sqrt{2}}{2}\right) = -\frac{1}{4} (\ln(\sqrt{2}) - \ln(2)) = -\frac{1}{4} \left(\frac{1}{2} \ln(2) - \ln(2)\right) = -\frac{1}{4} \left(-\frac{1}{2} \ln(2)\right) = \frac{\ln(2)}{8}$

Example 5.5.4

Find $\int x^2 \sqrt[3]{x+1} \, dx$

This is another common type of substitution problem that doesn't follow the usual patterns. You can recognize these types of problems by the product of a polynomial (in this case x^2) and another polynomial to some exponent (in this case $\sqrt[3]{x+1}$).

 Solution

Let $u = x + 1$. Then $du = dx$. Note, that we also need something to plug into the x^2 term, so solving for x we have $x = u - 1$ so $x^2 = (u - 1)^2$. Then,

$$\begin{aligned} \int x^2 \sqrt[3]{x+1} \, dx &= \int (u-1)^2 \sqrt[3]{u} \, du \\ &= \int (u^2 - 2u + 1) \sqrt[3]{u} \, du \\ &= \int (u^2 u^{\frac{1}{3}} - 2u u^{\frac{1}{3}} + 1 u^{\frac{1}{3}}) \, du \\ &= \int u^{\frac{7}{3}} - 2u^{\frac{4}{3}} + u^{\frac{1}{3}} \, du \end{aligned}$$

Note: because we moved the polynomial with multiple terms to the integer exponent, we could manually expand and distribute. Now we only need to use power rule a few times. Integrating we get

$$\begin{aligned} \int u^{\frac{7}{3}} - 2u^{\frac{4}{3}} + u^{\frac{1}{3}} \, du &= \frac{3}{10} u^{\frac{10}{3}} - \frac{6}{7} u^{\frac{7}{3}} + \frac{3}{4} u^{\frac{4}{3}} + C \\ &= \frac{3}{10} (x+1)^{\frac{10}{3}} - \frac{6}{7} (x+1)^{\frac{7}{3}} + \frac{3}{4} (x+1)^{\frac{4}{3}} + C. \end{aligned}$$

5.5 Section Summary:

- We reviewed the fundamental theorem of calculus, linking the area under a curve to the antiderivative of the function.
- We learned about one solution method for integrals: making a substitution.
- We noticed how the bounds of the integral change when we make substitutions.

6. Applications of Integrals

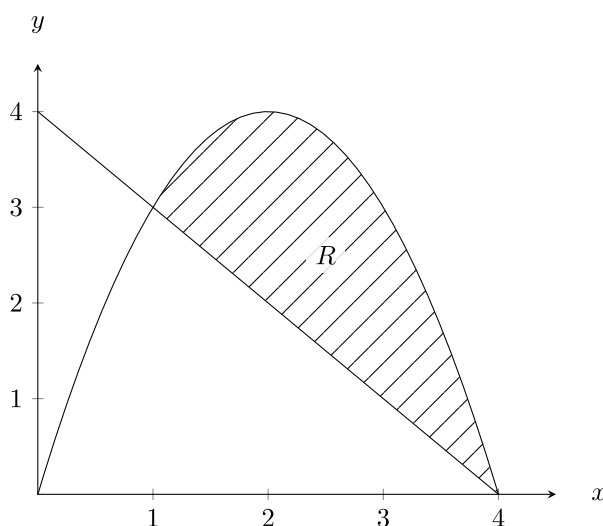
6.1. Area Between Curves

6.1 Learning Objectives

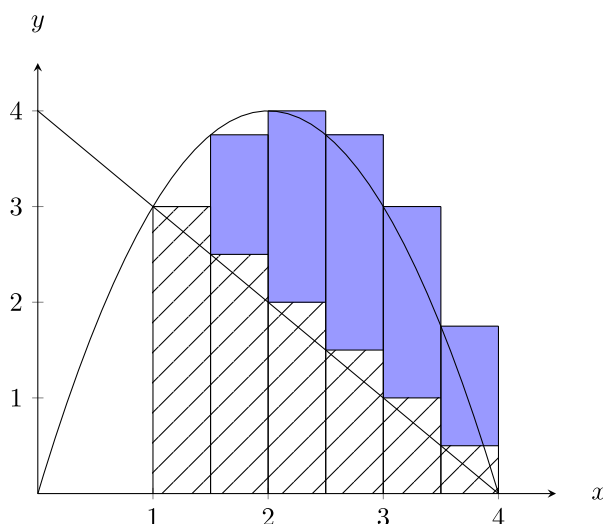
- I can extend the idea of area under a curve to area between curves.

Suppose we wish to find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.

How can we approximate this area?



To approximate this area, we can draw rectangles between the curves, with bottom edge lined up with the lower curve, and top edge lined up with the upper curve. This is effectively the same as estimating the area under each curve, and taking the difference between them (with the same number of rectangles). Note in the following plot, the first blue rectangle is the same height as the dashed one (because the functions intersect at the left side of this sub-interval), so it is not displayed.



We can approximate the area under both curves individually. Then, since the area under the “lower” curve is under both curves, we can subtract it away to get the area that is left.

Definition 6.1.1 Area Between Curves

Let R denote the region between continuous functions $y = f(x)$ and $y = g(x)$ and between the vertical lines $x = a$ and $x = b$, given that $f(x) \geq g(x)$ on $[a, b]$. The area of R is defined by

$$\begin{aligned} A &= \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x \right) - \left(\lim_{n \rightarrow \infty} \sum_{i=1}^n g(x_i^*) \Delta x \right) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (f(x_i^*) - g(x_i^*)) \Delta x. \end{aligned}$$

where we divide R into n strips of equal width, $\Delta x = \frac{b-a}{n}$ and height $f(x_i^*) - g(x_i^*)$. Thus, written as a definite integral, $A = \int_a^b (f(x) - g(x)) \, dx$.

Note

It isn't really necessary to define the area between curves in terms of Reimann sums. So long as you can reason out why the area between curves is the integral of their difference, you are good to go. You do also need to know the limit of Reimann sums definition of the integral, so it is good practice to go over it again here in a slightly more complicated format. Discuss in office hours if you have questions about this formulation.

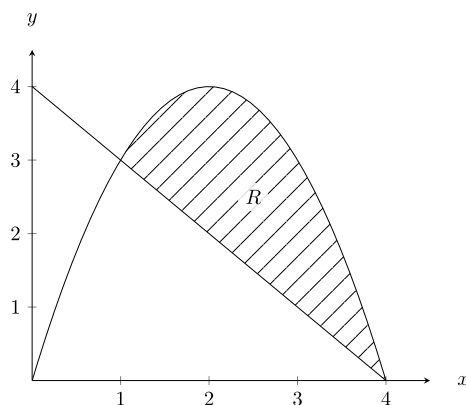
Exercise 6.1.1

Find the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[1, 4]$.

Solution

- It is good to sketch the region in these cases, and find any intersection points. Remember that we only get the area between curves if the upper function is, well, the upper function (we require $f(x) \geq g(x)$ on $[a, b]$ in the area between curves definition). If the functions changed which one was greater than the other over our interval, we would just break up the interval and compute a few integrals.

The region R was graphed before but is reproduced here for completion:



Note, on $[1, 4]$ we always have $-x^2 + 4x \geq -x + 4$. Then, the area, A , between curves is

$$\begin{aligned}
A &= \int_1^4 (-x^2 + 4x) - (-x + 4) \, dx \\
&= \int_1^4 -x^2 + 5x - 4 \, dx \\
&= \left[-\frac{1}{3}x^3 + \frac{5}{2}x^2 - 4x \right]_1^4 \\
&= -\frac{1}{3}(4)^3 + \frac{5}{2}(4)^2 - 4(4) - \left(-\frac{1}{3}(1)^3 + \frac{5}{2}(1)^2 - 4(1) \right) \\
&= \frac{19}{2}
\end{aligned}$$

Exercise 6.1.2

1. Compute the area between the curves $y = -x^2 + 4x$ and $y = -x + 4$ on $[0, 2]$. *Note: you will need to compute this using 2 definite integrals, make sure to check that the upper function is the first one in your difference.*
2. Convince yourself that the area between curves definition works even when $g(x) \leq 0$. You should be finding the *unsigned* area when you do this, what does that tell you about the sign of your answer to a problem asking for the area between curves?

Instead of taking vertical rectangles to approximate the area between curves, we might look at horizontal rectangles to find area between curves. This is especially useful in cases where we have more of a “right” and “left” curve instead of an “upper” and “lower” curve, as in the following example.

Example 6.1.1

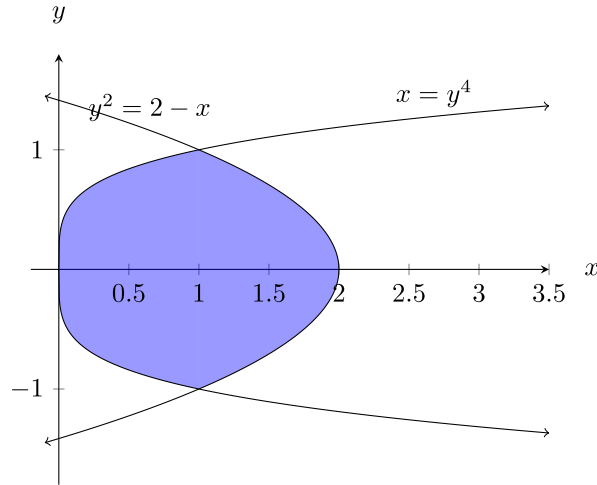
Set up and evaluate an integral to find the area enclosed by $x = y^4$ and $y^2 = 2 - x$.

Warning

It is expected that you know how to graph basic functions and similar relations like the ones above, although you may not have been formally trained in a class on these specifically. Please come discuss this in office hours if the hint to “turn your head sideways and think about swapping x and y ” does not help you. It is perfectly reasonable to be a little stuck on this when you first see it, and that is what office hours are there for!

Solution

First, we sketch the region to get an idea of what it looks like.



We see we have more of a right curve ($y^2 = 2 - x$) and left curve ($x = y^4$). In terms of x values, we have $x = y^4$ and $x = 2 - y^2$.

We start by finding where the functions intersect

$$\begin{aligned} y^4 &= 2 - y^2 \\ y^4 + y^2 - 2 &= 0 \\ (y^2 + 2)(y^2 - 1) &= 0 \end{aligned}$$

since $y^2 + 2$ is never 0, we know they intersect when $y^2 - 1 = 0$. To find the area between them, we can think about turning our head sideways and doing an integral in y , where instead of “upper function minus lower function” in the integral, we do “right function minus left function” (since $2 - y^2 \leq y^4$ for y on $[-1, 1]$ our method still applies). Thinking of this as an integral in y we have

$$\begin{aligned} \int_{-1}^1 (2 - y^2) - y^4 \, dy &= \left[2y - \frac{1}{3}y^3 - \frac{1}{5}y^5 \right]_{-1}^1 \\ &= 2 - \frac{1}{3} - \frac{1}{5} - \left(-2 + \frac{1}{3} + \frac{1}{5} \right) \\ &= \frac{44}{15} \end{aligned}$$

Example 6.1.2

Set up an integral to find the area enclosed between $f(x) = \cos(x)$ and $g(x) = \sin(x)$ on $[0, 2\pi]$.

Solution

Sketching the graphs on $[0, 2\pi]$ and using our knowledge of the unit circle, we know that the functions intersect at $x = \frac{\pi}{4}, \frac{5\pi}{4}$. For $[0, \frac{\pi}{4}]$, $f(x) \geq g(x)$ and then on $[\frac{\pi}{4}, \frac{5\pi}{4}]$, $g(x) \geq f(x)$, then at last on $[\frac{5\pi}{4}, 2\pi]$, $f(x) \geq g(x)$ once again. So, the area between the curves, A , is given by

$$A = \int_0^{\frac{\pi}{4}} (\cos(x) - \sin(x)) \, dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin(x) - \cos(x)) \, dx + \int_{\frac{5\pi}{4}}^{2\pi} (\cos(x) - \sin(x)) \, dx$$

 Tip

To find the area between 2 curves do the following

1. Identify the domain, so you have a good idea of what the region is.
2. Find where the curves intersect: set the equations of the curves equal to each other, and make sure to consider any domain restrictions.
3. Determine the subintervals over which you need to calculate the area: does the “upper curve” switch over the interval?
4. Determine whether $f(x) \geq g(x)$ or $g(x) \geq f(x)$ on each sub-interval.
5. Set up the integrals for each sub-interval with the greater function minus the smaller function (“upper minus lower” or “right minus left”) and evaluate.
6. Check: We are looking for *unsigned* area when we ask for the area between the curves, you should always get a positive number!

Exercise 6.1.3

Set up an integral to find the area of the triangular region enclosed by the points $(0, 0)$, $(2, 2)$, $(5, 0)$ with respect to both x and y .

6.1 Section Summary:

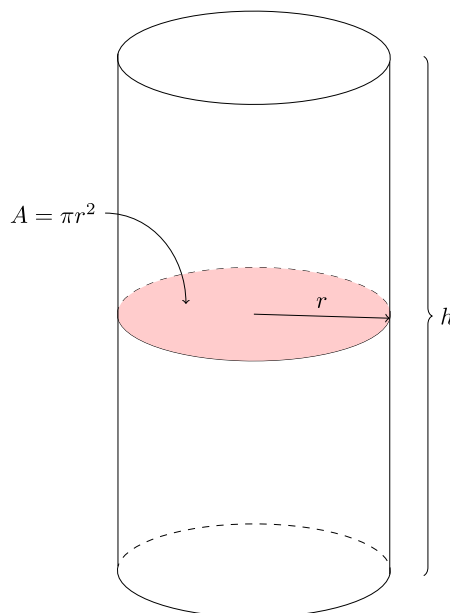
- We learned how to find the area between curves, either in the x or y direction.

6.2. Volumes by Cross-Sections, Disks, and Washers

6.2a Learning Objectives:

- I can think of *cross-sectional area* when calculating the volume of certain shapes using definite integrals.

Recall how we find the volume of a cylinder. $V = \pi r^2 h$. We essentially get the volume by taking the area of the base, πr^2 and sweeping it along the entire height h .

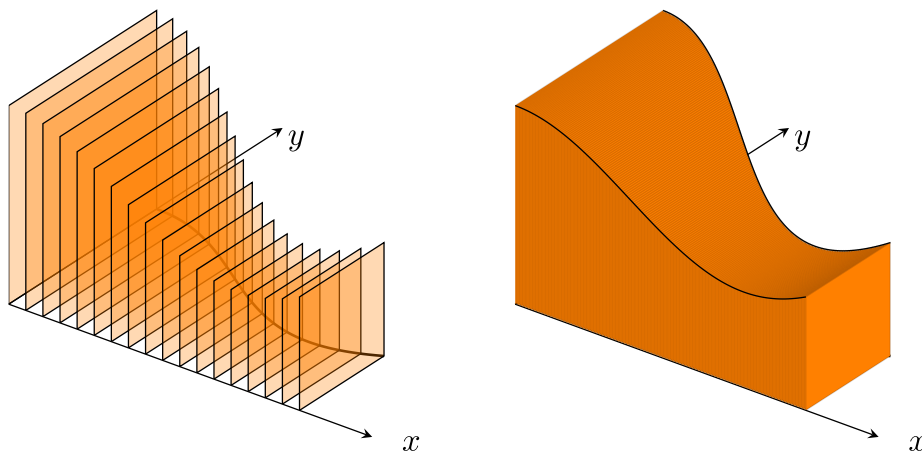


6.2.1. Slicing by Parallel Planes

Definition 6.2.1

We define a *cross-section* of a solid S as the plane region formed by intersecting S with a plane.

We can build a (3 dimensional) solid S by taking the same type of cross-section (e.g. squares) for each value of x , based on some region in the Cartesian plane.

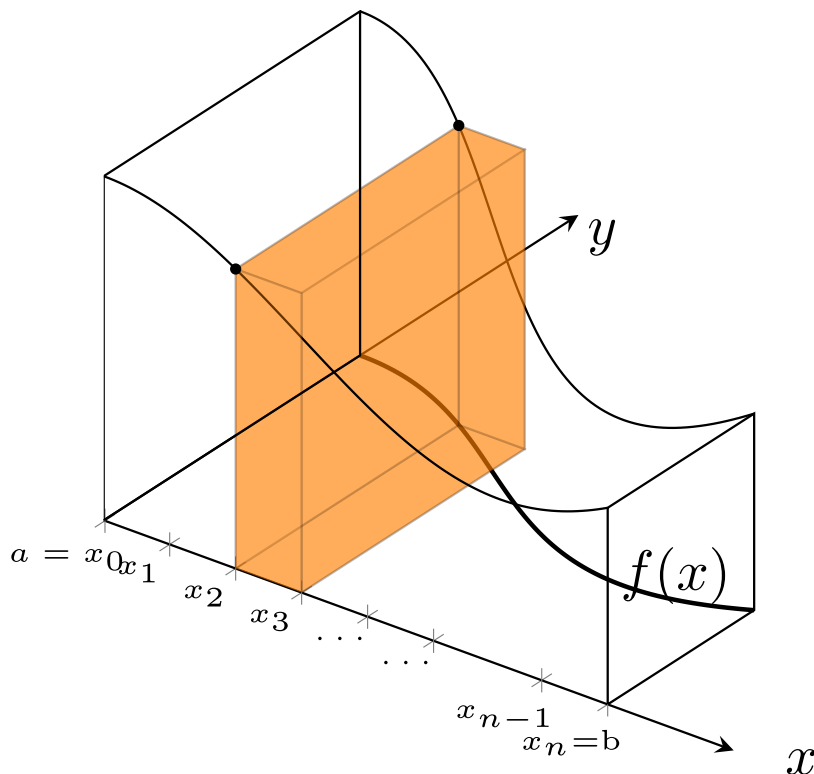


Suppose that we partition $[a, b]$ into subintervals

$$a = x_0 < x_1 < x_2 < x_3 < \dots < x_n = b$$

where each subinterval has width Δx . What is the volume of one slice of our solid? How would we use this to calculate the volume of the entire solid?

To approximate the volume, we can think of a cross-section with width Δx , then its volume is $V = (\text{Area of section})\Delta x$. Here the area of the cross-section is the area of a square with side length equal to the function value, $y = f(x)$.



As the width of the sub-intervals goes to zero and we add more rectangular prisms together, we get a better approximation of the volume. Once again, the limit of this sum ends up being a definite integral.

Definition 6.2.2

If the cross-section of a solid S at each point x in $[a, b]$ is a region $S(x)$ of area $A(x)$, where $A(x)$ is a continuous function of x , then

$$V = \int_a^b A(x) \, dx.$$

💡 Tip

To calculate the volume of a solid

1. Sketch a picture of the region and a typical cross-section.
 - Don't overcomplicate the cross-section, it should just be a prism with straight sides and top/bottom (or left/right sides) the same shape with area $A(x)$ (the area changes for each cross-section but is the same on the 2 faces of the single cross-section). It is never a part of a cone or pyramid or sphere,,
2. Find a formula for $A(x)$.

3. Find the limits of integration.

- Think about stretching the face with area $A(x)$ from one end of the shape to the other, where do you start/stop this process?

4. Integrate to find the volume, mind your units if applicable.

Note

Typical cross-sectional area formulae:

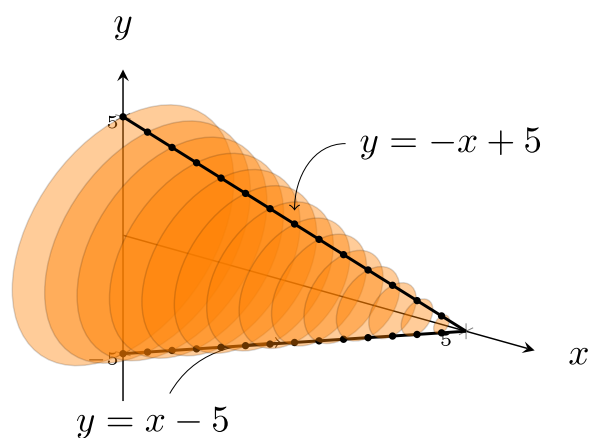
- Square: for a square with side length s , the area is $A = s^2$.
- Triangle: for a triangle with base b and height h , the area is $A = \frac{1}{2}bh$.
- Circle: for a circle with radius r , the area is $A = \pi r^2$.
- Semicircle: for a circle with radius r , the area is $A = \frac{1}{2}\pi r^2$.

Example 6.2.1

A cone with height 5 cm has a circular base with radius 5 cm. Set up a definite integral and evaluate it to find the volume of the cone.

Solution

A sketch of the volume, showing cross-sections, might look like:



We can think of slices of the cone being circles, with radius equal to the y value in $y = -x + 5$. This comes from the edge of the cone making the line $y = -x + 5$ in the x - y plane.

Then the area of one slice is $A(x) = \pi(-x + 5)^2$. The slices range from $x = 0$ to $x = 5$.

The definite integral we need to evaluate is

$$\begin{aligned}
 V &= \int_0^5 \pi(-x + 5)^2 dx \\
 &= \pi \int_0^5 x^2 - 10x + 25 dx \\
 &= \pi \left[\frac{1}{3}x^3 - 5x^2 + 25x \right]_0^5 \\
 &= \pi \left[\frac{1}{3}(5)^3 - 5(5)^2 + 25(5) \right] \\
 &= \frac{125}{3}\pi
 \end{aligned}$$

We are told the distances are measured in centimeters, so the final answer is the volume is $\frac{125\pi}{3}$ centimeters cubed.

There are 2 checks we can do at the end of the problem. First, our solution is positive, which we expect because volume is positive. Second, this is a cone and we do know the volume formula for a cone:

$$V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(5)^2(5) = \frac{125}{3}\pi.$$

This is the same thing we got from integrating, so our answer is correct!

Exercise 6.2.1

Derive the formula for the volume of a cone using similar techniques as above. *Hint: the line connecting the points $(0, r)$ and $(h, 0)$ will be useful.*

Example 6.2.2

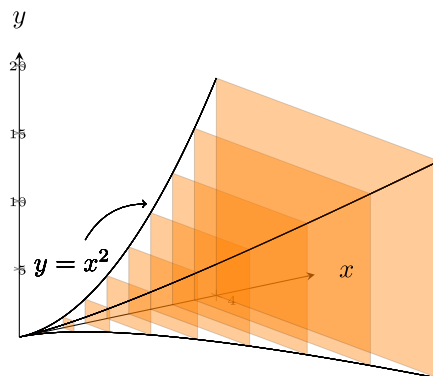
Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by $x = 0$, $x = 4$, and $y = x^2$. The cross-sections are perpendicular to the x -axis and are squares.

Set up an integral to find the volume of the given solid with the same base and square cross-sections, but the cross-sections are perpendicular to the y -axis.



Solution

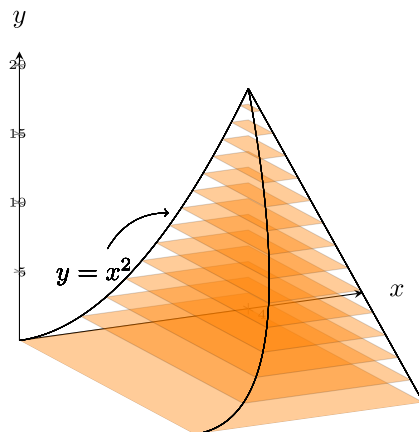
Sketch the region first, then



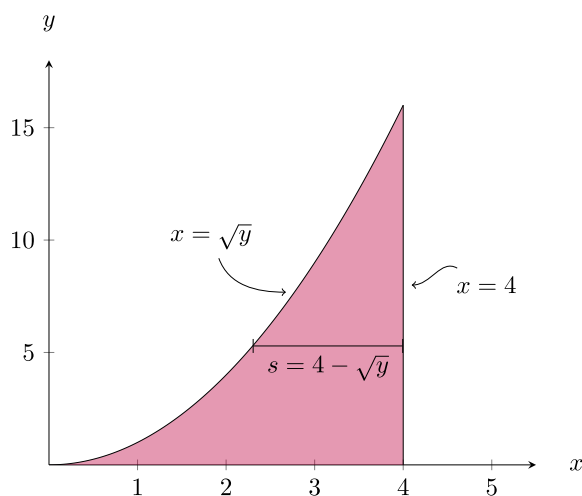
Where we see the side length of the square cross-sections is x^2 , and the slices fill the space from $x = 0$ to $x = 4$ so the definite integral we need to evaluate for the volume is

$$V = \int_0^4 (x^2)^2 dx = \int_0^4 x^4 dx.$$

For the second part of the problem, we are now looking at the region below:



Notice that the slices now go from bottom to top, when the cross-sections are perpendicular to y -axis, we want to use an integral in y . This region has square cross-sections where we can think about “right curve minus left curve” to find the side length of those squares:



Here the square side stretches between $x = \sqrt{y}$ and the vertical line $x = 4$. So, the side length is $s = 4 - \sqrt{y}$. This solid is formed up of slices from $y = 0$ to $y = 16$. Then the integral we solve for the volume is

$$V = \int_0^{16} (4 - \sqrt{y})^2 dy.$$

Exercise 6.2.2

Predict if you think the volumes of the two solids above will be the same or different. Then, evaluate each integral and compare. Were you correct or incorrect in your prediction? If you were incorrect, that is okay, but try to identify why/what went wrong.

Remark

In these problems, the variable of integration is determined based on whether the cross-sections are perpendicular to the x -axis or y -axis.

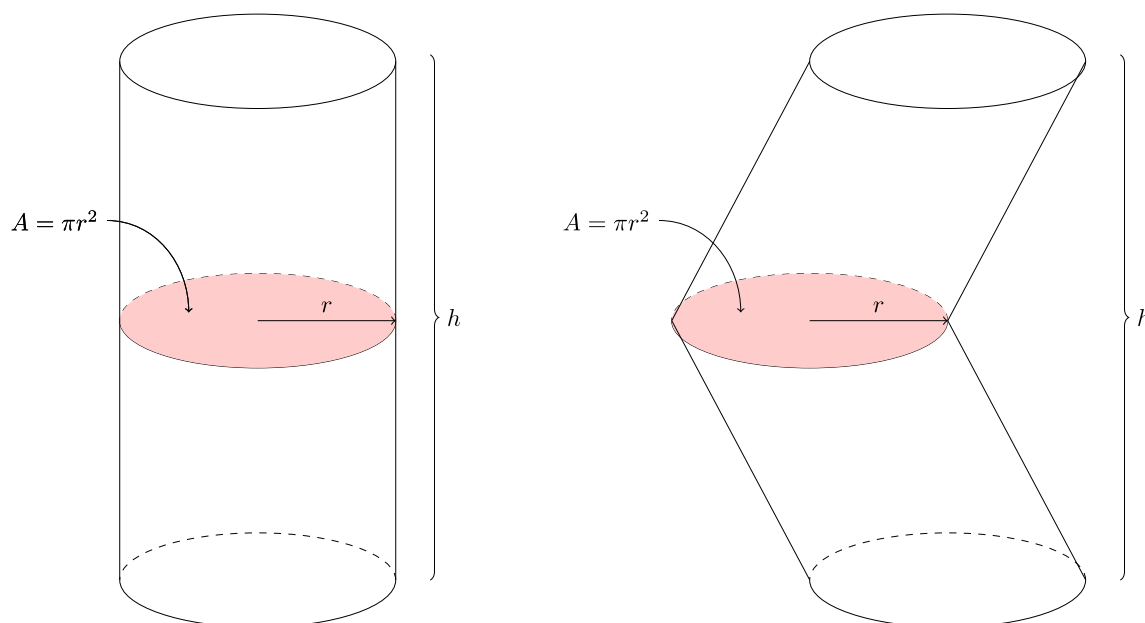
- Slices perpendicular to the x -axis (where slices stack from one x value to another) mean the variable of integration is x .
- Slices perpendicular to the y -axis (where slices stack from one y value to another) mean the variable of integration is y .

Property 6.2.1 Cavalieri's Principle

Solids with equal *height* and identical *cross-sectional area* at each height have the same volume.

Remark

These 2 cylinder-like solids have the same volume by Cavalieri's Principle



Exercise 6.2.3

Set up an integral to find the volume of the given solid. The base of the solid is the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$. The cross-sections perpendicular to the x -axis are equilateral triangles.

6.2a Section Summary:

- We found the area of cross-sections and set up integrals to find the volume of solids.

6.2.2. Solids of Revolution: The Disk method

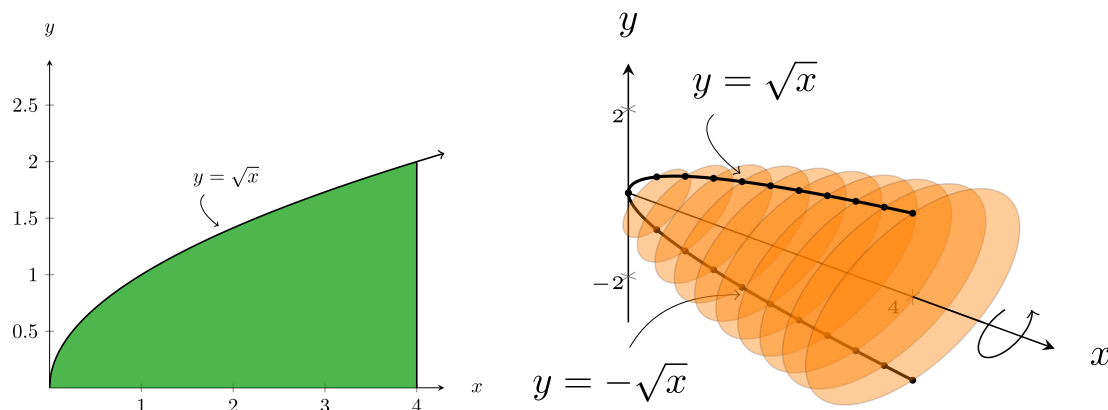
6.2b Learning Objectives:

- I can use disk or washer method to find the volume of a *solid of revolution*. I can recognize that these methods are related to finding the area between curves.

Definition 6.2.3

A *solid of revolution* is the solid generated by rotating (or revolving) a plane region about an axis in its plane.

Consider the function $y = \sqrt{x}$ on $[0, 4]$. Imagine rotating it all the way around the x -axis to form a solid:



Note

If we wish to find the volume of this solid, we have only added one step to our process from before. After that we have the same type of problem as the ones with the cross-sections. We have to build the 2-D region we want to form the solid from.

- Sketch the function and its reflection about the axis you are told to revolve around. This with the domain give you the 2-D region we go from.
- Imagine circular cross-sections that are perpendicular to the same axis you just used.
- Continuing as before the cross-sections will *always* be circular and have radius equal to the function value.

We can make this more specific below.

Theorem 6.2.1 Volume by Disks for Rotation about the x -axis

To find the volume of a solid formed by rotating a function $y = f(x)$ on $[a, b]$ about the x -axis first note that the area of each cross-section is $A(x) = \pi(f(x))^2$, then the volume is

$$V = \int_a^b A(x) \, dx = \int_a^b \pi(f(x))^2 \, dx.$$

Exercise 6.2.4

Write a similar statement to the one above for rotating a function $y = f(x)$ for y values in $[a, b]$ about the y -axis. *Hint: careful with the variables you use in the expression. The variable of integration should be the variable used in the integrand.*

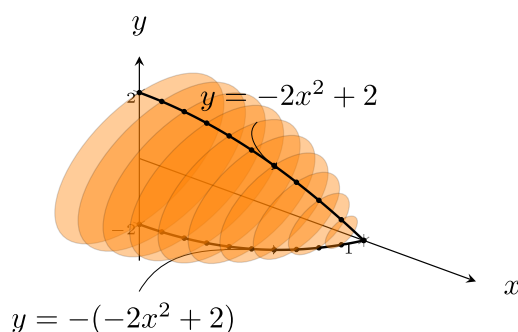
Example 6.2.3

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded above by $y = -2x^2 + 2$ around the x -axis.

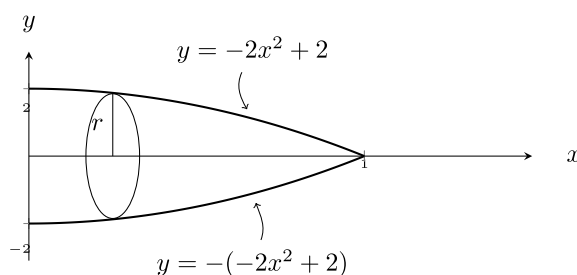
Set up an integral to find the volume when revolving this same region around the y -axis.

Solution

The solid looks like



However, this might be difficult to draw by hand. An alternative is to just draw the curve in 2-D, its reflection, and a single cross-section to help visualize the radius of the disks. This looks like



Then, we need to find where the curve hits the x -axis. Sometimes we can tell from the graph we draw, sometimes we have to solve for it

$$\begin{aligned} -2x^2 + 2 &= 0 \\ -2(x^2 - 1) &= 0 \\ -2(x + 1)(x - 1) &= 0 \\ x &= 1, -1 \end{aligned}$$

Since we are looking in the first quadrant, $x = 1$ is where this intersection happens (This is one of the bounds for the integral later). The area of the circular cross-section is $\pi r^2 = \pi(-2x^2 + 2)^2$. Thus, the integral we need is $V = \int_0^1 \pi(-2x^2 + 2)^2 dx$.

Exercise 6.2.5

Evaluate the integral above to find the volume of the solid.

Exercise 6.2.6

Set up and solve an integral to find the volume when revolving this same region around the y -axis.

Theorem 6.2.2 Volume by Disks for Rotation about the y -axis

To find the volume of a solid formed by rotating a function $y = f(x)$ (and solving for $x = f^{-1}(y)$) for x in $[a, b]$ and y between $f(a)$ and $f(b)$ about the y -axis first note that the area of each cross-section is $A(y) = \pi(f^{-1}(y))^2$, then the volume is

$$V = \int_{f(a)}^{f(b)} A(y) dy = \int_{f(a)}^{f(b)} \pi(f^{-1}(y))^2 dy.$$

In other words,

Note

Finding the Volume of a Solid of Revolution: Disk Method

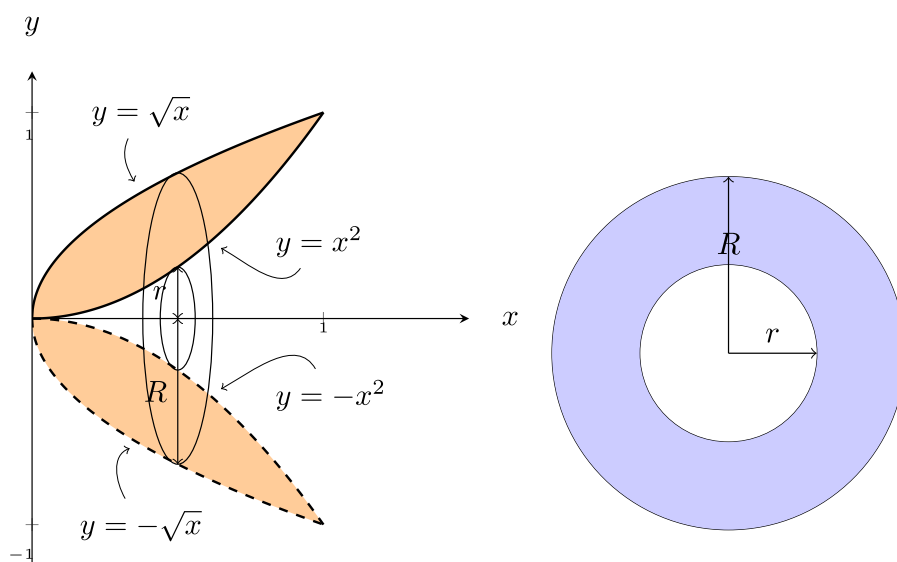
1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radius of the disk.
5. Integrate. Remember units in your final answer when applicable.

6.2.3. Solids of Revolution: The Washer Method

Note that we can only use the disk method if the cross-sections of the solid of revolution are circles. This means that there is no space gap between the region and the axis of revolution. What if there is a gap between the region and the axis of revolution?

Example 6.2.4

Set up an integral to find the volume of the solid generated by revolving the region in the first quadrant bounded by $y = x^2$ and $y = \sqrt{x}$ around the x -axis.



We note the curves intersect at $(1, 1)$. There are a few ways to think about this problem, but for now let's go back to thinking about the area of the cross-section and integrating that from 0 to 1 in x . For the cross-sections, the area is a circle with radius $R = \sqrt{x}$ minus the area of a smaller circle with area $r = x^2$. This region is sketched above in blue, and is known as a washer (as in the metal bit used with a nut and bolt that has the same shape). The area of the full region is the difference in areas between the 2 circles. That is, $A(x) = \pi R(x)^2 - \pi r(x)^2$. Then, integrating along the axis perpendicular to the cross-sectional area, we see the volume is

$$\begin{aligned} V &= \int_0^1 A(x) \, dx \\ &= \int_0^1 \pi R(x)^2 - \pi r(x)^2 \, dx \\ &= \pi \int_0^1 [(\sqrt{x})^2 - (x^2)^2] \, dx \\ &= \pi \int_0^1 x - x^4 \, dx \end{aligned}$$

Exercise 6.2.7

Evaluate the integral above.

Theorem 6.2.3 Volume by Washers for Rotation about the x -axis

For a solid of rotation formed by rotating a region around the x -axis where you have the required bounds and radii in terms of x , the volume is

$$V = \int_a^b A(x) \, dx = \pi \int_a^b [(R(x))^2 - (r(x))^2] \, dx$$

Theorem 6.2.4 Volume by Washers for Rotation about the y -axis

For a solid of rotation formed by rotating a region around the y -axis where you have the required bounds and radii in terms of y , the volume is

$$V = \int_a^b A(y) \, dy = \pi \int_a^b [(R(y))^2 - (r(y))^2] \, dy$$

In other words,

Note

Finding the Volume of a Solid of Revolution: Washer Method

1. Sketch both the region that is being rotated and the axis of revolution.
2. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
3. Determine the limits of integration. They must match the variable of integration!
4. Derive a formula for the radii of the washer: $R(x)$ and $r(x)$ (or $R(y)$ and $r(y)$ respectively).

- These are found by finding the distances of each curve from the axis of revolution. For more, see area between curves section.
5. Integrate. Remember units in your final answer when applicable.

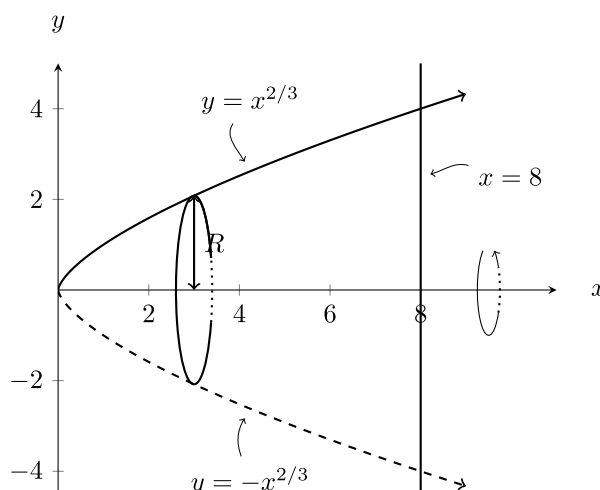
Example 6.2.5

Set up an integral to find the volume of the solid generated by revolving the region bounded by $y = x^{2/3}$, the x -axis, and $x = 8$

- around the x -axis
- around the y -axis
- around the line $y = 6$.

Solution

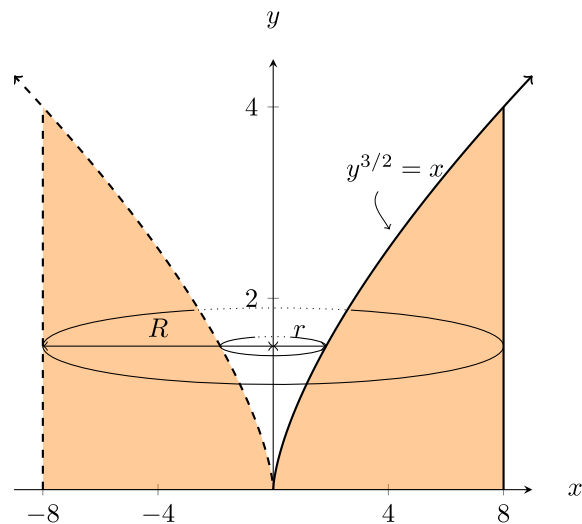
- Sketching the region and an idea of the volume we make by rotating around the x -axis we see the cross-sections are disks.



Thus, the volume can be found with the disk method: by thinking of the cross-sectional area as a bunch of circles perpendicular to the x -axis, from $x = 0$ to $x = 8$.

$$\begin{aligned}
 V &= \pi \int_0^8 (R(x))^2 dx \\
 &= \pi \int_0^8 \left(x^{2/3}\right)^2 dx \\
 &= \pi \int_0^8 x^{4/3} dx
 \end{aligned}$$

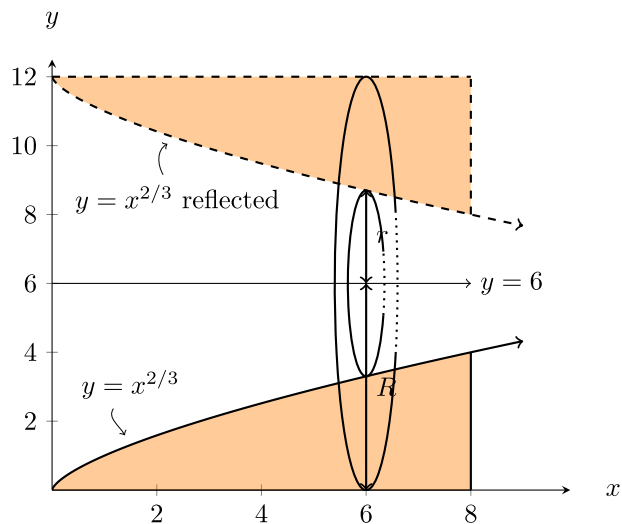
- Sketching the region and an idea of the volume we make by rotating around the y -axis we see the cross-sections are washers now.



Notice that we need to express the radii in terms of y now, the large radius is always $R(y) = 8$ and the small radius is $r(y) = y^{\frac{3}{2}}$ which we got from solving the original equation for x . Lastly, we note the solid fills space for y in $[0, 4]$. Then the volume integral will be

$$\begin{aligned} V &= \int_0^4 \pi [(R(y))^2 - (r(y))^2] dy \\ &= \int_0^4 \pi [(8)^2 - (y^{\frac{3}{2}})^2] dy \\ &= \pi \int_0^4 [64 - y^3] dy \end{aligned}$$

c) Sketching the region and the horizontal line $y = 6$ we have



We first find the radii, or the distance from $y = 6$ to the edges of the region. Note R is always 6, and $r = 6 - x^{\frac{2}{3}}$. Then the area of each washer as a function of x is

$$\begin{aligned} A(x) &= \pi R^2 - \pi r^2 \\ &= \pi [6^2 - (6 - x^{\frac{2}{3}})^2] \end{aligned}$$

The washers fill the space from $x = 0$ to $x = 8$, so the integral we use is

$$V = \pi \int_0^8 \left[6^2 - \left(6 - x^{\frac{2}{3}} \right)^2 \right] dx$$

6.2b Section Summary:

- We used the area of a cross-section to develop disk and washer method for finding volumes of the solids of rotation.
- We learned how to find the volumes of solids rotated about different horizontal and vertical lines.

6.3. Volumes by Cylindrical Shells

6.3 Learning Objectives:

- I can use shell method to find the volume of solids of rotation.
- I can identify scenarios when shell or washer method is required for a problem based on its geometry.

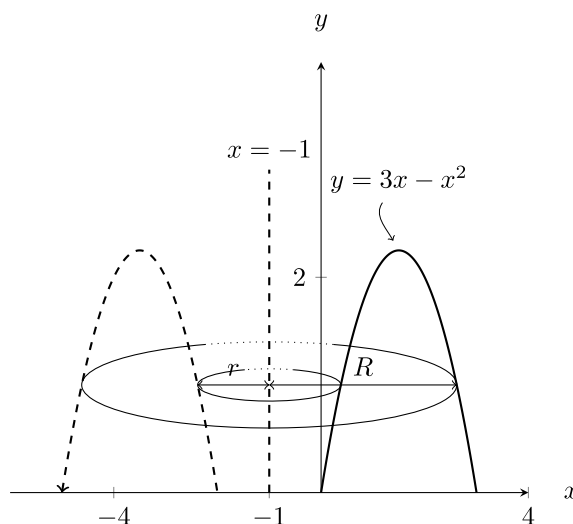
Exercise 6.3.1

Spend 5-10 minutes trying to set up the integral for using washer method for the following problem. Then stop and analyze why this is particularly difficult.

Rotate the region enclosed by the x -axis and the parabola $y = 3x - x^2$ around the line $x = -1$.

Solution

Note the plot of this region below



helps illustrate the difficulty. Simply put: We have no easy way of determining the radii, r and R , they are both the same function.

In the disk and washer method, the cross-sections taken were circular and perpendicular to the axis of rotation. We will introduce a new method, using a new shape for cross-sections, which are taken parallel to the axis of rotation.

Consider the following cookie cutters.



For the next method, we will think of finding the area of concentric shells making up the shape, then integrating from the center of the shells to the edge to “add” up all the areas that go into the volume.

Theorem 6.3.1 Volume by Cylindrical Shells (for revolutions about vertical lines)

The formula of the solid generated by revolving the region bounded by the x -axis, and the continuous function $y = f(x) \geq 0$ where $L \leq a \leq x \leq b$ about the vertical line $x = L$, is

$$V = \int_a^b 2\pi(\text{radius}) f(x) \, dx.$$

where $\text{radius} = |L - x|$ has been simplified

In general, the volume of the solid of a revolution sliced into cylindrical shells of radius, $r(x)$, and height, $h(x)$, is

$$V = \int_a^b 2\pi |L - x| h(x) \, dx.$$

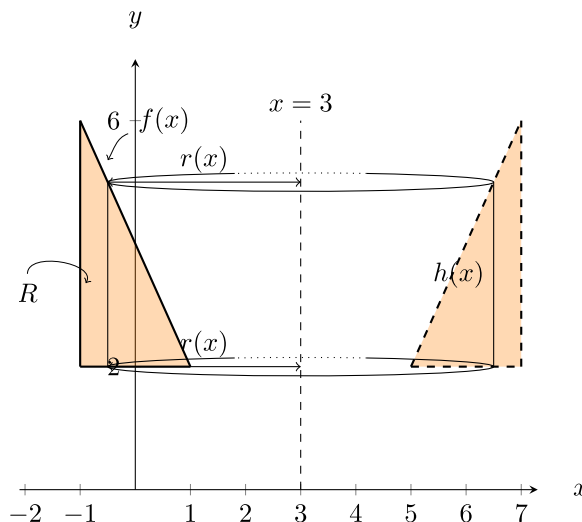
To justify the previous formula, think about the area of one of the cookie cutters, or shells.

Example 6.3.1

Set up and evaluate the integral for the volume of the solid obtained by rotating the region R about $x = 3$ where R is the region enclosed by $f(x) = 4 - 2x$, $x = -1$, and $y = 2$. Use shell method.

 Solution

- The axis of rotation is perpendicular to x -axis so we are integrating with respect to x for shell method.
- Sketching the region we have



- h is top curve minus bottom curve in this one, so $h(x) = 4 - 2x - 2 = 2 - 2x$
- $r = |L - x| = |3 - x|$, and since the bounds go from -1 to 1 this is the same as $r = 3 - x$.
- The bounds (listed above) can be found from finding the left-most and right-most x values of the region R .

Altogether, the volume is

$$\begin{aligned} V &= \int_a^b 2\pi r(x)h(x) \, dx \\ &= \int_{-1}^1 2\pi(3-x)(2-2x) \, dx \end{aligned}$$

To evaluate we multiply everything out so we can use power rule

$$\begin{aligned} V &= \int_{-1}^1 2\pi(3-x)(2-2x) \, dx \\ &= 2\pi \int_{-1}^1 6 - 6x - 2x + 2x^2 \, dx \\ &= 2\pi \int_{-1}^1 6 - 8x + 2x^2 \, dx \\ &= 2\pi \left[6x - 4x^2 + \frac{2}{3}x^3 \right]_{-1}^1 \\ &= 2\pi \left[\left(6 - 4 + \frac{2}{3} \right) - \left(-6 - 4 - \frac{2}{3} \right) \right] \\ &= \frac{80}{3}\pi \end{aligned}$$

Theorem 6.3.2 Volume by Cylindrical Shells (for revolutions about horizontal lines)

The formula of the solid generated by revolving the region bounded by the y -axis, and the continuous function $x = g(y) \geq 0$ where $L \leq a \leq y \leq b$ about the horizontal line $y = L$, is

$$V = \int_a^b 2\pi(\text{radius})g(y) \, dy.$$

where radius = $|L - y|$ has been simplified

In general, the volume of the solid of a revolution sliced into cylindrical shells of radius, $r(y)$, and height, $h(y)$, is

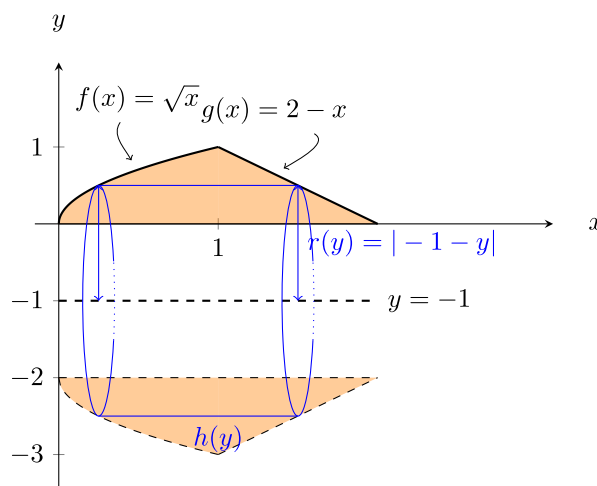
$$V = \int_a^b 2\pi |L - y| h(y) \, dy.$$

Example 6.3.2

Set up the integral of the volume of the solid obtained by revolving the region R enclosed by $f(x) = \sqrt{x}$, $g(x) = 2 - x$, and the x -axis about $y = -1$. (There may be more than one way to solve this problem, try shell now and see if washer works later. Compare and contrast both)

 Solution

First we sketch the region, R , and it's reflection across $y = -1$. Then we can start to sketch out the shell.



Since we are rotating about an axis parallel to the x -axis, we will integrate with respect to y (for shell method). We find the functions intersect at $(1, 1)$. To find $h(y)$, we think about right function minus left function, and we need to write these in terms of y . The right curve is given by $x = 2 - y$. The left curve, $f(x)$, is $x = y^2$. So the height of each shell is $h(y) = 2 - y - y^2$. The radius is $r = |L - y| = y + 1$. The region R has y values from 0 to 1. Altogether,

$$\begin{aligned}
 V &= \int_0^1 2\pi r(y)h(y) \, dy \\
 &= \int_0^1 2\pi(y+1)(2-y-y^2) \, dy
 \end{aligned}$$

Exercise 6.3.2

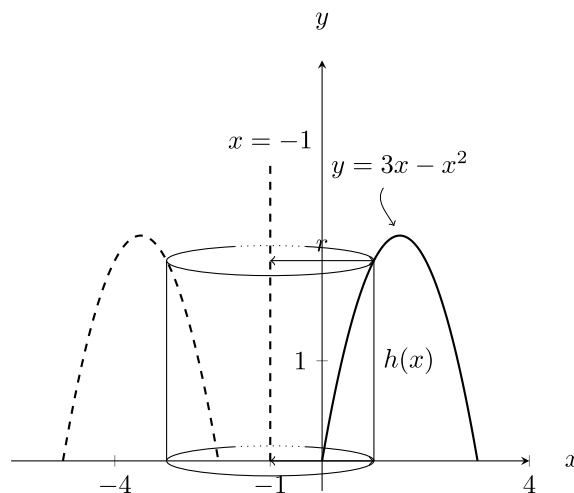
Use washer method to set up the integral for the above problem now. Practice writing both integrals as a limit of Riemann sums.

Example 6.3.3

Set up an integral to find the volume of the following solid: Rotate the region enclosed by the x -axis and the parabola $y = 3x - x^2$ around the line $x = -1$.

Solution

Our shells look like



Then, the height is $h(x) = 3x - x^2$. Notice that since we did not revolve around the y -axis, that the radius is not just x . Instead, the radius is the distance between x and $x = -1$. From area between curves we can think about “right curve minus left curve” to see that $r(x) = x - (-1) = x + 1$. The integral for the volume is then

$$V = \int_0^3 2\pi(x+1)(3x-x^2) \, dx$$

Tip

Finding the Volume of a Solid of Revolution: Shell method

1. Sketch the region that is being rotated.
2. Draw the axis of rotation.

3. Determine the variable of integration. Integration is performed with respect to the axis that is perpendicular to the slices.
4. Derive a formula for the height of the shell. The shell height can be found by drawing a line parallel to the axis of rotation. This might be top curve minus bottom curve, or right curve minus left curve. Write this in terms of the variable of integration from the previous step.
 - $|L - x|$ or $|L - y|$ depending on variable of integration.
5. Determine the bounds of integration. They must match the variable of integration!
6. Integrate. Remember units in your final answer when applicable.
 - $V = \int_a^b 2\pi r h \, dx$ or maybe with respect to y

Exercise 6.3.3

Practice working out/evaluating all the unfinished integrals in this section. This is good practice, but not necessarily part of this section in a way.

6.3 Section Summary:

- We learned when and how to use shell method to find the volume of solids of rotation.

6.4. Work

6.4 a Learning Objectives:

- I can understand the definitions of work and force.
- I understand how to use integration to describe work.
- I can describe the work done in pulley systems and springs.

Definition 6.4.1

Force is any influence that causes an object to undergo a change in speed, direction, or shape. Force has magnitude and direction. We say that an object with mass, m , and acceleration, a , has force

$$F = ma$$

A body moves a distance, d , along a straight line after being acted by a force of constant magnitude, F , in the direction of motion, we define the work, W , done by the force on the body with the formula

$$W = Fd$$

6.4.1. Units

In this class you will encounter both metric system and imperial system.

In metric,

- Distance should be converted to meters, m, to work with other units.
- Mass should be converted to kilograms, kg, to work with other units.
- Time is measured in seconds, s.
- Force is measured in Newtons, N, where 1 N is equivalent to 1 kg m s^{-2} .
- Work is measured in Joules, J, where 1 J is equivalent to 1 N m.
- Acceleration due to gravity is $g = 9.8 \text{ m s}^{-2}$.

In imperial system,

- Distance is measured in feet, ft
- Mass is not normally given. Instead weight (which is a force) is given.
- Force is measured in pounds, lb.
- Work is measured in foot-pounds, ft-lb.

Example 6.4.1

Determine the amount of work done.

- a. A 5 kg book is lifted 4 m.
- b. A 5 lb book is lifted 4 ft.

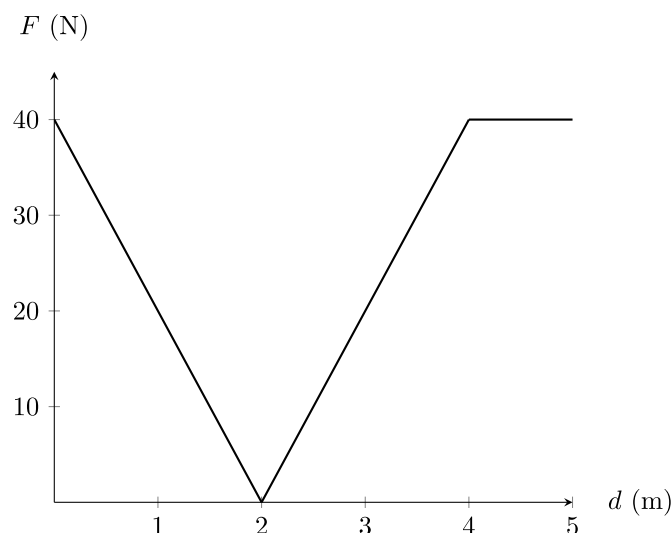
Solution

- a. $W = mad = (5 \text{ kg}) (9.8 \text{ m s}^{-2}) (4 \text{ m}) = 196 \text{ J}$.
- b. $W = Fd = (5 \text{ lb}) (4 \text{ ft}) = 20 \text{ ft-lb}$.

6.4.2. Non-Constant Force

Example 6.4.2

The graph of a force function (in Newtons) is shown below. How much work is done by the force in moving an object a distance of 5 m?



Solution

$$\begin{aligned} W &= Fd \\ &= \text{area under curve} \\ &= A_1 + A_2 + A_3 \end{aligned}$$

Then $A_1 = 2(40)\left(\frac{1}{2}\right) = 40$ J, $A_2 = 2(40)\left(\frac{1}{2}\right) = 40$ J, and $A_3 = 1(40) = 40$ J.

Altogether, the work is

$$\begin{aligned} W &= A_1 + A_2 + A_3 \\ &= 120 \text{ J} \\ &= \int_0^5 F(x) \, dx \end{aligned}$$

6.4.3. Lifting Objects

Example 6.4.3

A uniform cable 30 ft long and weighing 60 lb hangs vertically from a pulley system at the top of a building. A steel beam weighs 500 lb and is attached to the end of the cable. Set up and evaluate an integral to find the work required to pull the beam to the top. Explicitly write out the integral as a limit of Riemann sums.

Solution

We will consider the cable and the steel beam separately. First, the work to get the steel beam to the top of the building is $W_B = Fd = 500(30) = 15\,000$ ft-lb.

As we pull the cable up, less of it is pulling back, so the force is changing as we lift more and more cable. To account for this, consider chopping the cable up into n tiny segments of length $\Delta x = \frac{30}{n}$ ft. The i th segment is x_i ft from the top of the building. The weight (force) of each segment is the cable density times the length of the segment:

$$\begin{aligned} F_i &= \frac{60 \text{ lb}}{30 \text{ ft}} \Delta x \\ &= 2\Delta x \text{ lbs} \end{aligned}$$

The work done to lift the i th segment of cable to the top of the building is then $W_i = F_i x_i = 2\Delta x x_i$.

Next, we add up the work for all n segments to get an approximation for the total work done to lift the cable to the top of the building:

$$\begin{aligned} W_C &\approx \sum_{i=1}^n W_i \\ &\approx \sum_{i=1}^n 2\Delta x x_i \end{aligned}$$

To get the exact amount of work, we take the limit as the number of cable segments goes to infinity,

$$\begin{aligned} W_c &= \lim_{n \rightarrow \infty} \sum_{i=1}^n 2\Delta x x_i \\ &= \int_0^{30} 2x \, dx \end{aligned}$$

The work for the cable is then $W_C = \int_0^{30} 2x \, dx = x^2|_0^{30} = 900$ ft-lb.

Combined with the beam, the total work done to lift the beam and cable to the top of the building is $900 + 15000 = 15\,900$ ft-lb.

Exercise 6.4.1

Assume the previous example's set up. Find the work done to lift the beam only 10 ft up.

6.4.4. Stretching and Compressing Springs

Theorem 6.4.1 Hooke's Law

The force required to compress or stretch a spring by x units short of or beyond its natural length is proportional to x

$$F = kx$$

Note that k is called the spring constant.

Exercise 6.4.2

Using what you know about the units for force and distance, derive the units for the spring constant in both metric and imperial units.

Example 6.4.4

A spring has a natural length of 50 cm. A force of 50 N stretches the spring to a length of 60 cm.

1. What force is needed to stretch the spring x meters?
2. How long is the spring when stretched by a force of 200 N?
3. Set up an integral to find the work done in stretching the spring 20 cm.
4. Set up an integral to find the work done in stretching the spring from a length of 100 cm to 110 cm.
5. 20 J of work is done in stretching the spring a certain distance. How far was the spring stretched?

Solution

1. In general, $F = kx$, so if we know k we can get a formula for the force needed to stretch the spring x meters. Going from 50 cm to 60 cm is a change of 0.1 m, since this takes 50 N we know

$$\begin{aligned}50 \text{ N} &= k(0.1 \text{ m}) \\k &= \frac{50 \text{ N}}{0.1 \text{ m}} \\k &= 500 \text{ N m}^{-1}\end{aligned}$$

Then a formula for force needed to stretch this spring x meters is $F = 500x$.

2. Using our formula and $F = 200$,

$$\begin{aligned}F &= 500x \\200 &= 500x \\x &= 0.4 \text{ m}\end{aligned}$$

Remember that this is how far from the spring rest length it is, so we have to add the 0.5 m to this to get that the spring is 0.9 m.

3. 20 cm is 0.2 m. Then the work is

$$\begin{aligned}W &= \int_a^b F(x) \, dx \\&= \int_0^{0.2} 500x \, dx\end{aligned}$$

4. The trick here is finding the bounds of the integral. 100 cm is 1 m and 110 cm is 1.1 m. We also need to know how far from rest length these distances are: $1 - 0.5 = 0.5$ m and $1.1 - 0.5 = 0.6$ m. Then, the work done is

$$W = \int_{0.5}^{0.6} 500x \, dx$$

5. Let l be how far the spring was stretched (from rest). Then the work to get it there is

$$W = \int_0^l 500x \, dx = 20 \text{ J}$$

Then solving for l

$$20 = \int_0^l 500x \, dx$$

$$20 = 250x^2 \Big|_0^l$$

$$20 = 250l^2$$

$$l^2 = \frac{2}{25}$$

$$l = \pm \frac{\sqrt{2}}{5}$$

The spring was stretched an additional $\frac{\sqrt{2}}{5}$ m.

6.4 a Section Summary:

- We used integrals and Riemann sums to describe work done to lift heavy objects and stretch springs.

6.4 b Learning Objectives:

- I can understand the definitions of work and force.
- I understand how to use integration to describe work.
- I can describe the work done when pumping fluids out of containers of varying shapes.

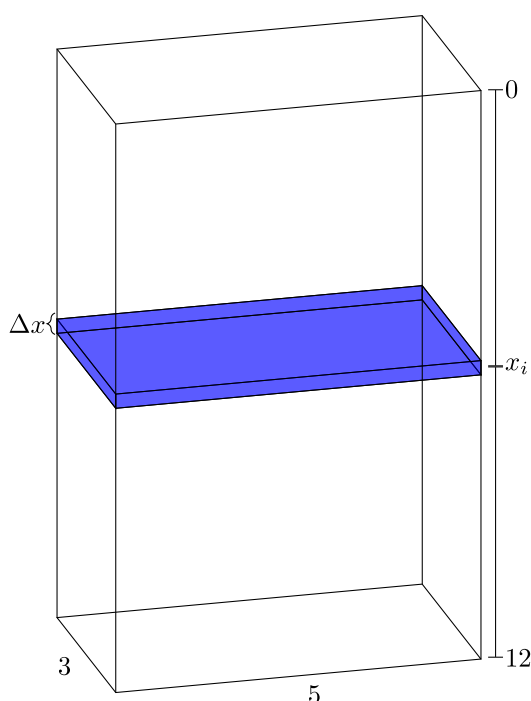
Another physical system we can calculate the work done in is when pumping liquids out of a tank.

6.4.5. Pumping Liquids

Example 6.4.5

A rectangular tank full of water has height 12 ft, width 3 ft, and length 5 ft. Find the work done in pumping all of the water to the top of the tank given that the density of water is 62.4 lbs ft^{-3} .

Solution



Consider pumping a single layer of water out at a time. The work done to pump the i th layer of water out is $W_i = F_i d_i$. We will say the i th layer has to be pumped x_i ft out of the tank, and x_i goes from 0 to 12.

Then the force on each layer is the weight of the slice which can be found by multiplying density and volume: $F_i = 62.4 \text{ lbs ft}^{-3} V_i$. Now $V_i = (3)(5)(\Delta x)$, where Δx is the height of the slice. Altogether $F_i = 62.4(15)\Delta x$ lb.

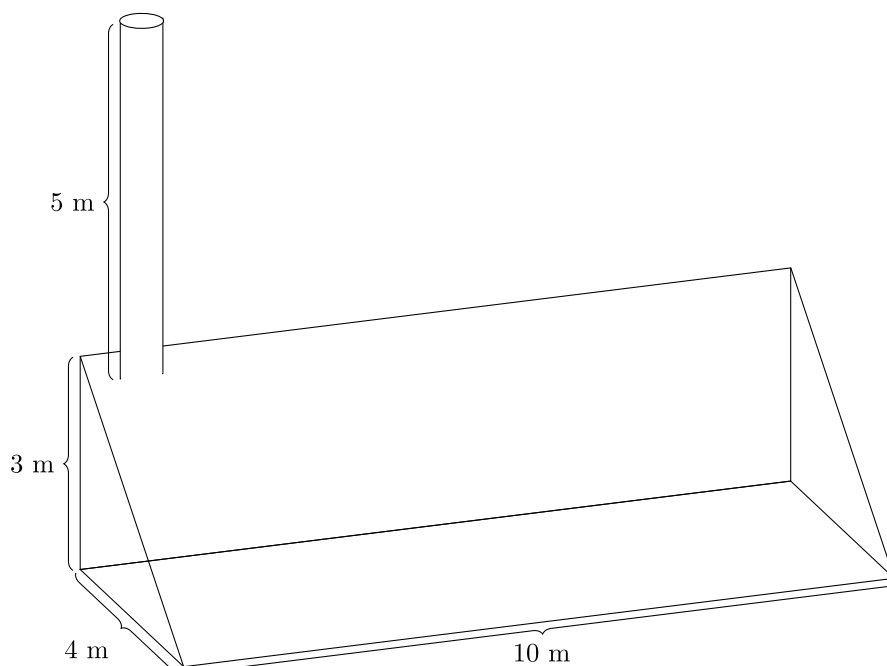
Then the work on each slice is $W_i = 62.4(15)\Delta x x_i$ lbft.

The work is then the limit of the sums of the layers

$$\begin{aligned}
 W &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (62.4)(15) \Delta x_i \\
 &= \int_0^{12} (15)(62.4)x \, dx
 \end{aligned}$$

Example 6.4.6

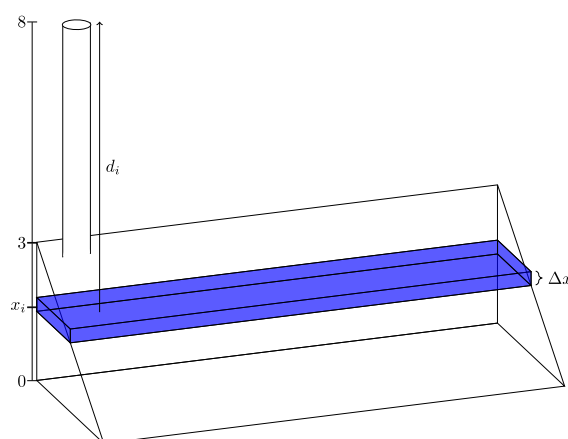
A tank pictured below is full of water. The density of water is 1000 kg m^{-3} . Set up the integral to find the work needed to pump the water to a height of 5 m above the top of the tank.



Solution

Recall that to find the work done, we consider the work to pump a generic layer of water out, then add all the layers up and take the limit as the number of layers goes to infinity to get an integral.

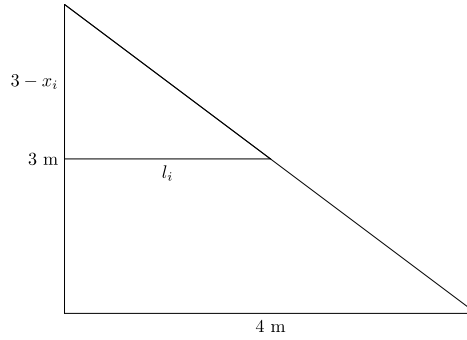
First, we sketch the layer



Then, the force on this layer is just gravity, so

$$\begin{aligned} F_i &= m_i a \\ &= (\text{density of water})(V_i)(g) \\ &= (1000 \text{ kg m}^{-3})(V_i)(9.8 \text{ m s}^{-2}) \end{aligned}$$

The volume of the slice is a little trickier, we need to use similar triangles to find the side length, l_i , of the box that touches the triangular face in terms of x_i .



We get the following relation for l_i .

$$\begin{aligned} \frac{4}{3} &= \frac{l_i}{3 - x_i} \\ l_i &= \frac{4}{3}(3 - x_i) \end{aligned}$$

Altogether the volume is

$$\begin{aligned} V_i &= l_i w h \\ &= \frac{4}{3}(3 - x_i)(10)\Delta x \end{aligned}$$

So the force to lift this layer becomes

$$F_i = (1000 \text{ kg m}^{-3}) \frac{4}{3}(3 - x_i)(10)\Delta x(9.8 \text{ m s}^{-2})$$

Then the distance this layer has to travel to get out of the spout at the top is $d_i = 8 - x_i$, where x_i falls between $[0, 3]$ (since that is where the water is).

Thus, the work to pump a single layer is

$$W_i = (1000 \text{ kg m}^{-3}) \frac{4}{3}(3 - x_i)(10)\Delta x(9.8 \text{ m s}^{-2})(8 - x_i).$$

The sum of the work for each layer gives us an approximation to the total work

$$W \approx \sum_{i=1}^n (1000 \text{ kg m}^{-3}) \frac{4}{3}(3 - x_i)(10)\Delta x(9.8 \text{ m s}^{-2})(8 - x_i).$$

Then taking the limit of this sum gives us the integral for work

$$\begin{aligned}
 W &= \lim_{n \rightarrow \infty} \sum_{i=1}^n (1000) \frac{4}{3} (3 - x_i) (10) \Delta x (9.8) (8 - x_i) \\
 &= \int_0^3 (1000) \frac{4}{3} (3 - x) (10) (9.8) (8 - x) \, dx \text{ J.}
 \end{aligned}$$

If you want to evaluate this integral, bring constant multiples out of the integral and multiply the binomials in x then use power rule.

6.4 b Section Summary:

- We used limits of Riemann sums to describe the work done to pump liquids out of tanks.
- We had to review some formulas for some geometry to finish these problems, like similar triangles and Pythagorean Theorem.

6.5. Average Value of a Function

6.5 Learning Objectives:

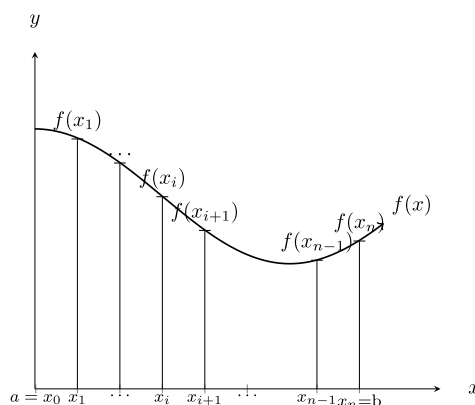
- I can find the average value of a function on some interval.
- I can know and understand how to apply the Mean Value Theorem.

Definition 6.5.1

Recall the average of n values, namely $a_1, a_2, \dots, a_n$, is

$$\text{average} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} = \sum_{i=1}^n \frac{1}{n} a_i$$

Suppose we wanted to find the average of all values that a function $f(x)$ achieves on an interval $[a, b]$. We can think about sampling the function at n equally spaced points and dividing by the number of points. As we look at sampling more points, we get a more precise approximation for the average of f .



The approximate function average is

$$\begin{aligned} f_{\text{ave}} &\approx \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n} \\ &= \frac{1}{n} \sum_{i=1}^n f(x_i) \\ &= \frac{\Delta x}{b-a} \sum_{i=1}^n f(x_i) \\ &= \frac{1}{b-a} \sum_{i=1}^n f(x_i) \Delta x \end{aligned}$$

To make our approximation more accurate, we will sample the function at even more points, eventually taking the limit gives us the integral.

$$\begin{aligned} f_{\text{ave}} &= \frac{1}{b-a} \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\ &= \frac{1}{b-a} \int_a^b f(x) \, dx. \end{aligned}$$

Definition 6.5.2

The average value of f on $[a, b]$ is given by

$$f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

Example 6.5.1

Find the average of $f(x) = 1 + x^2$ on $[-1, 2]$.

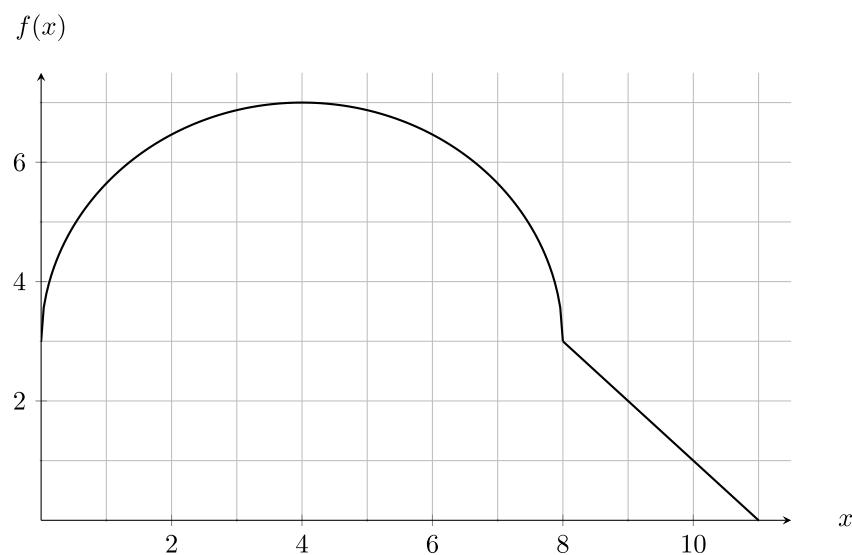
 Solution

We use the formula

$$\begin{aligned} f_{\text{ave}} &= \frac{1}{2 - (-1)} \int_{-1}^2 1 + x^2 \, dx \\ &= \frac{1}{3} \left[x + \frac{x^3}{3} \right]_{-1}^2 \\ &= \frac{1}{3} \left[2 + \frac{8}{3} - \left(-1 - \frac{1}{3} \right) \right] \\ &= 2 \end{aligned}$$

Example 6.5.2

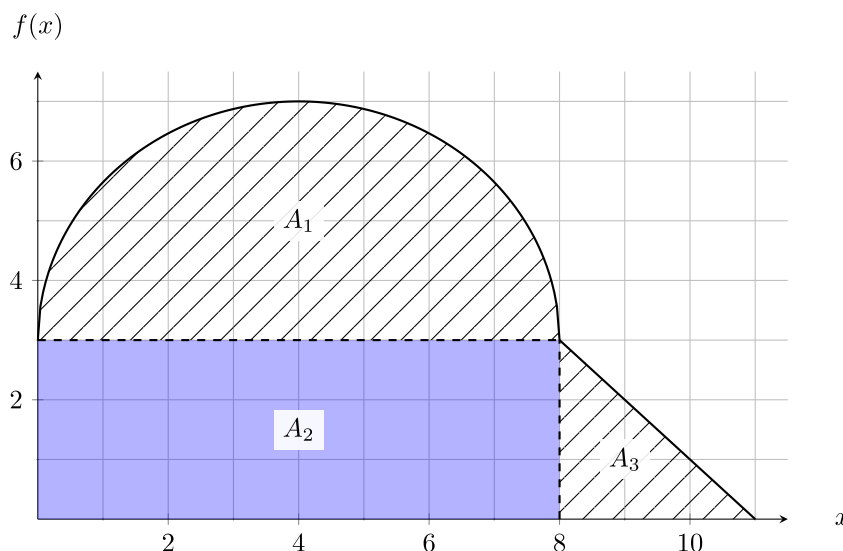
Find the average value of f , given in the graph below, on $[0, 11]$.

 **Solution**

We use the formula for average of a function

$$f_{\text{ave}} = \frac{1}{11-0} \int_0^{11} f(x) \, dx$$

In this case, to get the integral, we remember that the integral of a curve is the same as the area under that curve. We don't need the equation for f since we can use geometry to find the area.



Now, we add the areas of the different parts up: $A_1 = \frac{1}{2}\pi r^2 = \frac{1}{2}\pi 4^2 = 8\pi$ And $A_2 = wh = (8)(3) = 24$. And $A_3 = \frac{1}{2}bh = \frac{1}{2}(3)(3) = \frac{9}{2}$.

Then, the average is

$$\begin{aligned} f_{\text{ave}} &= \frac{1}{11} \int_0^{11} f(x) \, dx \\ &= \frac{1}{11} (A_1 + A_2 + A_3) \\ &= \frac{1}{11} \left(8\pi + 24 + \frac{9}{2} \right). \end{aligned}$$

Theorem 6.5.1 Mean Value Theorem for Integrals

If f is continuous on $[a, b]$, then there is some c in (a, b) where

$$f(c) = f_{\text{ave}} = \frac{1}{b-a} \int_a^b f(x) \, dx$$

Example 6.5.3

Does the function $f(x) = (x+1)^3$ achieve its average value on $[a, b]$ for some value in (a, b) ? If so where does this happen?

 Solution

Yes, because f is continuous (since it is a polynomial), so by Mean Value Theorem (MVT) the function achieves its average value for some c in (a, b) .

As for where this happens, the MVT does not say. We can find it though,

$$\begin{aligned} f_{\text{ave}} &= \frac{1}{2-0} \int_0^2 (x+1)^3 \, dx \\ &= \frac{1}{2} \int_1^3 u^3 \, du \\ &= \frac{1}{2} \frac{u^4}{4} \Big|_1^3 \\ &= \frac{1}{2} \left(\frac{3^4}{4} - \frac{1}{4} \right) \\ &= 10 \end{aligned}$$

Where the u -sub we used was $u = x + 1$ so $du = dx$ and $u(0) = 1$ and $u(2) = 3$.

Since the average value is now known (10), we can find which c value has $f(c) = 10$.

$$\begin{aligned} 10 &= (c+1)^3 \\ \sqrt[3]{10} &= c+1 \\ \sqrt[3]{10} - 1 &= c \end{aligned}$$

which is in $(0, 2)$.

6.5 Section Summary:

- We found the average value of a function on some interval.
- We used mean value theorem to find where a function achieves its average value.

7. Integration Techniques

7.1. Integration by Parts

7.1a Learning Objectives

- I understand how integration by parts compares to the product rule.
- I understand when and how to use integration by parts.
- I can choose an appropriate u and v .
- I can use integration by parts to reduce integrals.

Recall that u -substitution corresponded to “undoing” the chain rule (when taking derivatives). What corresponds to the product rule?

Product Rule:

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Integrating both sides, and rearranging slightly, we end up with the following

$$\begin{aligned}\frac{d}{dx}[f(x)g(x)] &= f(x)g'(x) + f'(x)g(x) \\ f(x)g(x) + C &= \int f(x)g'(x) \, dx + \int f'(x)g(x) \, dx \\ f(x)g(x) - \int f'(x)g(x) \, dx &= \int f(x)g'(x) \, dx\end{aligned}$$

Relabeling the last line with a few substitutions $u = f(x) \Rightarrow du = f'(x) \, dx$ and $v = g(x) \Rightarrow dv = g'(x) \, dx$ we end up with

Theorem 7.1.1 Integration by Parts

$$\int u \, dv = uv - \int v \, du$$

Example 7.1.1

Find the antiderivative:

$$\int x \sin(x) \, dx$$

Solution

We don't know a best method for picking u and dv yet, so let's try something and see what happens. Let

$$\begin{aligned}
 u &= \sin(x), & dv &= x \, dx, \\
 du &= \cos(x) \, dx, & v &= \int x \, dx \\
 & & &= \frac{x^2}{2}
 \end{aligned}$$

Then

$$\int x \sin(x) \, dx = \sin(x) \frac{x^2}{2} - \underbrace{\int \frac{x^2}{2} \cos(x) \, dx}_{\text{this integral looks worse}}$$

Try the other way around now

$$\begin{aligned}
 u &= x, & dv &= \sin(x), \\
 du &= dx, & v &= \int \sin(x) \\
 & & &= -\cos(x)
 \end{aligned}$$

This gives us

$$\begin{aligned}
 \int x \sin(x) \, dx &= -x \cos(x) - \underbrace{\int -\cos(x) \, dx}_{\text{which we know}} \\
 &= -x \cos(x) + \sin(x) + C
 \end{aligned}$$

Exercise 7.1.1

Check the above answer by taking the derivative. Did you get what we start with?

7.1.1. How to we choose which function to use for u ?

i Note

Recall that u is a function of which we can easily derive, but not necessarily integrate. We tend to pick things that simplify a lot when we take the derivative of them: logarithms have $\frac{d}{dx} \log(x, b) = \frac{1}{\ln(b)x}$, inverse trig functions have become almost algebraic like $\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1-x^2}}$, and polynomials which decrease in number of term and power the more you derive them. Alternatively, things that don't change very much like trig functions and exponential functions might be avoided for u .

Here is a way to discern potential functions for u , keeping in mind that you often have to try a few things as you get used to the process.

1. **I**: Inverse trig functions
2. **L**: logarithmic functions
3. **A**: algebraic functions (polynomials, rational functions)
4. **T**: trig functions
5. **E**: exponential functions

7.1.2. Indefinite Integrals

Example 7.1.2

Find

$$\int 2xe^x \, dx.$$

Solution

If we consider taking the exponential term as our choice for u , then when we take the derivative of this term, it stays exactly the same. On the other hand, if we let $u = 2x$ then when we take the derivative we get something simpler. This leads us to want to try $u = 2x$ first. (But we might have to try something else later)

Then,

$$\begin{aligned} u &= 2x, & dv &= e^x \, dx, \\ du &= 2 \, dx, & v &= \int e^x \, dx = e^x \end{aligned}$$

gives us

$$\begin{aligned} \int 2xe^x \, dx &= 2xe^x - \int 2e^x \, dx \\ &= 2xe^x - 2e^x + C \end{aligned}$$

Example 7.1.3

Find

$$\int e^{\sqrt{x}} \, dx$$

Solution

This problem is a little tricky, it needs a substitution first, and then it can be integrated by parts. Let $w = \sqrt{x}$ then to get dx we can solve for x : $w^2 = x$ so $2w \, dw = dx$. Then the integral becomes

$$\int e^{\sqrt{x}} \, dx = \int 2we^w \, dw$$

Which we solved in the previous problem. So,

$$\int e^{\sqrt{x}} \, dx = 2\sqrt{x}e^{\sqrt{x}} - 2e^{\sqrt{x}} + C$$

Exercise 7.1.2

Find

$$\int x^2 e^x \, dx.$$

Tip

Hint: Try something.

7.1a Section Summary:

- We learned how to “undo” the product rule, via integration by parts.
- We learned to use substitution and integration by parts in the same problem.
- We learned that sometimes you have to do integration by parts multiple times to get a final answer.

7.1.3. Definite Integrals

7.1b Learning Objectives

- I can use integration by parts to solve definite integrals.
- I understand when and how to use integration by parts.
- I can choose an appropriate u and v .
- I can use integration by parts to reduce integrals.

Recall the product rule once more to see what we do when the integrals have bounds.

Product Rule:

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Integrating both sides from a to b , and rearranging slightly, we end up with the following

$$\begin{aligned}\frac{d}{dx}[f(x)g(x)] &= f(x)g'(x) + f'(x)g(x) \\ f(x)g(x)|_a^b &= \int_a^b f(x)g'(x) dx + \int_a^b f'(x)g(x) dx \\ f(x)g(x)|_a^b - \int_a^b f'(x)g(x) dx &= \int_a^b f(x)g'(x) dx\end{aligned}$$

Relabeling the last line with a few substitutions $u = f(x) \Rightarrow du = f'(x) dx$ and $v = g(x) \Rightarrow dv = g'(x) dx$ we end up with

Theorem 7.1.2 Integration by Parts for Definite Integrals

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

Note

Note that in practice, we still have the same variable of integration before and after applying integration by parts. We aren't really substituting anything since the variables u and v don't stick around. This way, the bounds are still for the original variable of integration and shouldn't be changed. This is different from the substitution method, where the variable of integration does change, so the bounds must also change.

Example 7.1.4

Find

$$\int_0^1 x3^x dx.$$

 Solution

$$\begin{aligned}u &= x, \quad dv = 3^x dx, \\du &= dx, \quad v = \int 3^x dx \\&= \frac{1}{\ln(3)} 3^x\end{aligned}$$

Then we have

$$\begin{aligned}\int_0^1 x 3^x dx &= \left[\frac{x}{\ln(3)} 3^x \right]_0^1 - \int_0^1 \frac{1}{\ln(3)} 3^x dx \\&= \left[\frac{1}{\ln(3)} 3 - 0 \right] - \left[\frac{1}{(\ln(3))^2} 3^x \right]_0^1 \\&= \frac{3}{\ln(3)} - \left[\frac{3}{(\ln(3))^2} - \frac{1}{(\ln(3))^2} \right] \\&= \frac{3}{\ln(3)} - \frac{2}{(\ln(3))^2}\end{aligned}$$

Example 7.1.5

Find

$$\int_0^1 \arctan(x) dx.$$

 Solution

We want to pick $u = \arctan(x)$ since inverse trig functions get simpler (for integrating) after we take the derivative. Notice that this just leaves $dv = dx$. That is okay, and can sometimes be helpful even!

$$\begin{aligned}u &= \arctan(x), \quad dv = dx, \\du &= \frac{1}{1+x^2} dx, \quad v = x\end{aligned}$$

Then

$$\begin{aligned}\int_0^1 \arctan(x) dx &= [x \arctan(x)]_0^1 - \int_0^1 \frac{x}{1+x^2} dx \\&= \frac{\pi}{4} - 0 - \int_0^1 \frac{x}{1+x^2} dx\end{aligned}$$

Now we just need to evaluate

$$\int_0^1 \frac{x}{1+x^2} dx$$

Take a second to finish this on your own before checking the solution below. You know how to do this.

We need a substitution, since we see a function and its derivative: Let $w = 1 + x^2$. Then $dw = 2x \, dx$. Since this is an actual substitution, we need to change the bounds as well $w(0) = 1$ and $w(1) = 2$. Lastly, if you want to call this u instead of w , that is fine too. I just picked something to avoid confusion with integration by parts. Then

$$\begin{aligned}\int_0^1 \frac{x}{1+x^2} \, dx &= \frac{1}{2} \int_1^2 \frac{1}{w} \, dw \\ &= \frac{1}{2} [\ln|w|]_1^2 \\ &= \frac{1}{2} \ln(2)\end{aligned}$$

Altogether

$$\int_0^1 \arctan(x) \, dx = \frac{\pi}{4} - \frac{1}{2} \ln(2).$$

Exercise 7.1.3

Find

$$\int_3^5 \ln(x) \, dx.$$

Sometimes you have to use integration by parts multiple times to solve an integral. This happens with powers of trig functions, “double transcendental functions”, and some other things.

Example 7.1.6

Find

$$\int e^x \cos(x) \, dx.$$

When thinking about what to pick for u here, if we pick either e^x or $\cos(x)$ we run into similar problems: nothing ever really changes.

We can take advantage of the fact that nothing really changes by some really slick algebra shown in the following solution.

Solution

Let’s give the thing we want to find a name: Let $I = \int e^x \cos(x) \, dx$.

Now, apply integration by parts a few times (picking the same type of thing for u) until you see I show up again, like so

$$\begin{aligned}u &= \cos(x), & dv &= e^x \, dx, \\ du &= -\sin(x) \, dx, & v &= e^x\end{aligned}$$

$$\begin{aligned}
 I &= \int e^x \cos(x) \, dx \\
 &= \cos(x)e^x - \underbrace{\int e^x(-\sin(x)) \, dx}_{\neq I}
 \end{aligned}$$

Since the thing in the integral is not I , we repeat on this new integral term.

$$\begin{aligned}
 u &= -\sin(x), & dv &= e^x \, dx, \\
 du &= -\cos(x) \, dx, & v &= e^x
 \end{aligned}$$

Then

$$\begin{aligned}
 I &= \int e^x \cos(x) \, dx \\
 &= \cos(x)e^x - \left[\int e^x(-\sin(x)) \, dx \right] \\
 &= \cos(x)e^x - \left[-\sin(x)e^x - \int e^x(-\cos(x)) \, dx \right] \\
 &= \cos(x)e^x + \sin(x)e^x - \underbrace{\int e^x \cos(x) \, dx}_{=I}
 \end{aligned}$$

So we end up with the algebraic equation which we can solve for the answer, I :

$$\begin{aligned}
 I &= \cos(x)e^x + \sin(x)e^x - I \\
 2I &= \cos(x)e^x + \sin(x)e^x \\
 I &= \frac{1}{2}[\cos(x)e^x + \sin(x)e^x]
 \end{aligned}$$

Exercise 7.1.4

Given $n \geq 2$ is an integer, prove the **reduction formula**:

$$\int (\sin(x))^n \, dx = -\frac{1}{n} \cos(x) \sin^{n-1}(x) + \frac{n-1}{n} \int \sin^{n-2}(x) \, dx$$

Alternatively, use integration by parts to find $\int \sin^2(x) \, dx$. Hint use $u = \sin(x)$ for the first integration by parts step and follow similar steps as the previous problem.

If you want, we can go over this in office hours.

7.1b Section Summary:

- We learned how to handle bounds on integrals when we need to use integration by parts.
- We also looked at some examples that need multiple steps of integration by parts to solve.

7.2. Trigonometric Integrals

7.2 Learning Objectives

- I combine methods from previous sections to find general strategies to solve trigonometric integrals.

Example 7.2.1

$$\int \cos^3(x) \, dx$$

Note

With our current techniques, we can try to recognize what function has a derivative of this, try a u -substitution, or integration by parts (or reduction formula). We don't clearly know this antiderivative, and we can't use u -sub because there isn't a spot for du . Lastly, we could try integrating by parts and/or reduction formula, but that will be incredibly tedious. Let's try the alternative in the solution below.

Whenever we see a power of $\cos(x)$ (or $\sin(x)$) of 2 or more we can try to use the Pythagorean Identity:

Theorem 7.2.1 Pythagorean Identity

For any angle x we have

$$\sin^2(x) + \cos^2(x) = 1$$

This can be rearranged by dividing by $\sin^2(x)$ or $\cos^2(x)$:

$$1 + \cot^2(x) = \csc^2(x)$$

and

$$\tan^2(x) + 1 = \sec^2(x).$$

Solution

Recall the Pythagorean Identity $\sin^2(x) + \cos^2(x) = 1$, rearranging this we can write $\cos^2(x) = 1 - \sin^2(x)$. Then using this in the integral we have

$$\begin{aligned} \int \cos^3(x) \, dx &= \int \cos^2(x) \cos(x) \, dx \\ &= \int (1 - \sin^2(x)) \cos(x) \, dx \end{aligned}$$

This might look more complicated at first, but now if we tried a u -sub it will work! Let $u = \sin(x)$, then $du = \cos(x) \, dx$, so the integral becomes

$$\begin{aligned}
 \int (1 - \sin^2(x)) \cos(x) \, dx &= \int 1 - u^2 \, du \\
 &= u - \frac{u^3}{3} \\
 &= \sin(x) - \frac{\sin^3(x)}{3}
 \end{aligned}$$

Let's check our answer by taking its derivative:

$$\begin{aligned}
 \frac{d}{dx} \sin(x) - \sin^3 \frac{x}{3} &= \cos(x) - \sin^2(x) \cos(x) \\
 &= \cos(x)(1 - \sin^2(x)) \\
 &= \cos(x) \cos^2(x) \\
 &= \cos^3(x)
 \end{aligned}$$

Note we used the same Pythagorean Identity here to get back to the same thing we started with.

Warning

In this section and on its WebAssign problems you may have to use other trigonometric identities. It is good to remember these exist every now and then, and practice using them, but having you memorize too many of these is not worth your time and energy.

You are expected to know the Pythagorean identity and alternative versions of it, but any other identity you need you may look up. Examples include the half angle and double angle formulas, and product to sum formulas, etc.

Notice in the above problem that in our u -sub, it was nice to have everything in terms of 1 trig function ($\sin(x)$ in the above example), except a single multiple of the other trig function times dx ($\cos(x)$ in the above example). Then the first variable can be our u . We will repeat this technique below, getting rid of most of one trig function except something to work for du . This takes a lot of practice and trial and error to get good at, make sure to try things and be okay failing and trying again.

Example 7.2.2

Find

$$\int_0^{\frac{\pi}{2}} \sin^5(x) \cos^2(x) \, dx.$$

Because we need 1 trig function left over for du , and the power of $\sin(x)$ is odd, we can leave 1 $\sin(x)$ alone, and replace the other 4 powers with $\cos(x)$ through the Pythagorean identity.

 Solution

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^5(x) \cos^2(x) \, dx &= \int_0^{\frac{\pi}{2}} \sin^4(x) \sin(x) \cos^2(x) \, dx \\ &= \int_0^{\frac{\pi}{2}} (1 - \cos^2(x))^2 \sin(x) \cos^2(x) \, dx\end{aligned}$$

And now a u -sub works! Let $u = \cos(x)$, then $du = -\sin(x) \, dx$, and $u(0) = 1$ and $u(\frac{\pi}{2}) = 0$.

So we have

$$\begin{aligned}\int_0^{\frac{\pi}{2}} (1 - \cos^2(x))^2 \sin(x) \cos^2(x) \, dx &= \int_1^0 (1 - u^2)^2 u^2 (-1) \, du \\ &= \int_0^1 (1 - u^2)^2 u^2 \, du \\ &= \int_0^1 u^2 - 2u^4 + u^6 \, du \\ &= \left[\frac{u^3}{3} - \frac{2}{5}u^5 + \frac{u^7}{7} \right]_0^1 \\ &= \frac{1}{3} - \frac{2}{5} + \frac{1}{7}\end{aligned}$$

Example 7.2.3

Find

$$\int \sin^2(x) \cos^2(x) \, dx.$$

Here, we have an even number on both exponents, so using the Pythagorean identity wouldn't leave anything for du at the end of the rearranging. We need a new strategy.

Theorem 7.2.2 Double Angle Formula

For any angle x we have

$$\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$$

and

$$\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$$

 Solution

When we use double angle formula, we might have to simplify a lot and try the formula again.

$$\begin{aligned}\int \sin^2(x) \cos^2(x) \, dx &= \int \frac{1}{2}(1 - \cos(2x)) \frac{1}{2}(1 + \cos(2x)) \, dx \\ &= \frac{1}{4} \int 1 - \cos^2(2x) \, dx\end{aligned}$$

Then we have to use double angle formula again.

$$\begin{aligned}\int \sin^2(x) \cos^2(x) \, dx &= \frac{1}{4} \int 1 - \cos^2(2x) \, dx \\ &= \frac{1}{4} \int 1 - \frac{1}{2}(1 + \cos(4x)) \, dx \\ &= \frac{1}{4} \int 1 - \frac{1}{2} - \cos(4x) \, dx\end{aligned}$$

Then let $u = 4x$, $du = 4 \, dx$ so

$$\begin{aligned}\int \sin^2(x) \cos^2(x) \, dx &= \frac{1}{4} \int 1 - \frac{1}{2} - \frac{\cos(4x)}{2} \, dx \\ &= \frac{1}{4} \int \frac{1}{2} \, dx - \frac{1}{32} \int \cos(u) \, du \\ &= \frac{1}{8}x - \frac{1}{32} \sin(4x) + C.\end{aligned}$$

Tip

Strategy for evaluating $\int \sin^m(x) \cos^n(x) \, dx$

- **Case 1: At least one of m or n are odd:**
 - Choose the trig function with the odd power to be u . If both are odd, just pick one.
 - Save one factor of this trig function for du later.
 - Rewrite the other trig function (even power) with the Pythagorean Identity.
 - Do u -sub, expand everything, and evaluate.
- **Case 2: Both m and n are even**
 - Reduce all even powers with the double angle identities.
 - You may need to expand and apply the previous step multiple times until all even powers are eliminated.
 - The identity $\sin(2\theta) = 2 \sin(\theta) \cos(\theta)$ may sometimes be useful.

Example 7.2.4

Find

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \tan^5(x) \sec^7(x) \, dx$$

Building off what we know from $\sin(x)$ and $\cos(x)$, we will probably want to pick u so that du is still around. Recall the derivatives of the 4 non sinusoidal trig functions

 Note

You should know the following derivatives from the prerequisites to this class, if you are unfamiliar with these, please review and/or discuss in office hours.

$$\frac{d}{dx} \tan(x) = \sec^2(x)$$

$$\frac{d}{dx} \cot(x) = -\csc^2(x)$$

$$\frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

$$\frac{d}{dx} \csc(x) = -\csc(x) \cot(x)$$

 Solution

Now, when we pick u we should plan to “save” the derivative of u and replace everything else so it is in terms of u . We can either keep a $\sec^2(x)$ around, or $\sec(x) \tan(x)$. Depending on if u is going to be $\tan(x)$ or $\sec(x)$. Either way should work, we will pick $u = \sec(x)$ here, try $u = \tan(x)$ on your own as practice.

Let $u = \sec(x)$, then $du = \sec(x) \tan(x) dx$. This gets us

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \tan^4(x) \underbrace{\sec^6(x)}_{u^6} \underbrace{\tan(x) \sec(x) dx}_{du}$$

This leaves us with $\tan^4(x)$ that we have to rewrite with Pythagorean identity. Then, using $\tan^2(x) = \sec^2(x) - 1$ we have

$$\begin{aligned} \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (\sec^2(x) - 1)^2 \sec^6(x) \tan(x) \sec(x) dx &= \int_{\frac{2}{\sqrt{3}}}^{\frac{2}{\sqrt{2}}} (u^2 - 1)^2 u^6 du \\ &= \int_{\frac{2}{\sqrt{3}}}^{\frac{2}{\sqrt{2}}} u^{10} - 2u^8 + 1u^6 du \\ &= \left[\frac{u^{11}}{11} - \frac{2}{9}u^9 + \frac{u^7}{7} \right]_{\frac{2}{\sqrt{3}}}^{\frac{2}{\sqrt{2}}} \end{aligned}$$

Simplifying this expression is left as an exercise to the reader.

Example 7.2.5

Find

$$\int \cot^2(x) \csc^4(x) dx$$

 Solution

Let $u = \cot(x)$, then $du = -\csc^2(x) dx$. Then,

$$\int \cot^2(x) \csc^4(x) dx = \int (-1) \underbrace{\cot^2(x)}_{u^2} \underbrace{\csc^2(x)}_{\text{rewrite}} \underbrace{(-1) \csc^2(x) dx}_{du}$$

Then using the Pythagorean Identity $\csc^2(x) = 1 + \cot^2(x)$ we have

$$\begin{aligned} \int \cot^2(x) \csc^4(x) dx &= - \int \cot^2(x) (1 + \cot^2(x))^2 (-\csc^2(x)) dx \\ &= - \int u^2 (1 + u^2)^2 du \\ &= - \int u^2 + 2u^4 + u^6 du \\ &= - \left[\frac{u^3}{3} + \frac{2}{5} u^5 + \frac{u^7}{7} \right] + C \\ &= - \left[\frac{\cot^3(x)}{3} + \frac{2}{5} \cot^5(x) + \frac{\cot^7(x)}{7} \right] + C \end{aligned}$$

 Tip

Strategy for evaluating $\int \tan^m(x) \sec^n(x) dx$

- **Case 1: n even:** (i.e. the power of secant is even)
 - Save one factor of $\sec^2(x)$ for du later.
 - Rewrite the rest of the powers of $\sec^2(x)$ using $\sec^2(x) = 1 + \tan^2(x)$ (even power) with the Pythagorean Identity.
 - Do u -sub, expand everything, and evaluate.
- **Case 2: m is odd:** (i.e. the power of tangent is odd)
 - Save one factor of $\sec(x) \tan(x)$ for du later.
 - Convert the remaining even powers of tangent to $\tan^2(x) = \sec^2(x) - 1$
 - Do u -sub, expand everything, and evaluate.
- You may also use the following 2 antiderivatives

$$\int \tan(u) du = \ln|\sec(u)| + C$$

$$\int \sec(u) du = \ln|\sec(u) + \tan(u)| + C$$

Exercise 7.2.1

Derive the above antiderivatives. The second needs a clever multiplication by 1 trick. Come to office hours if you want to go through this together.

7.2 Section Summary:

- We used trigonometric identities and other tools to rearrange trigonometric integrals to be able to solve them.

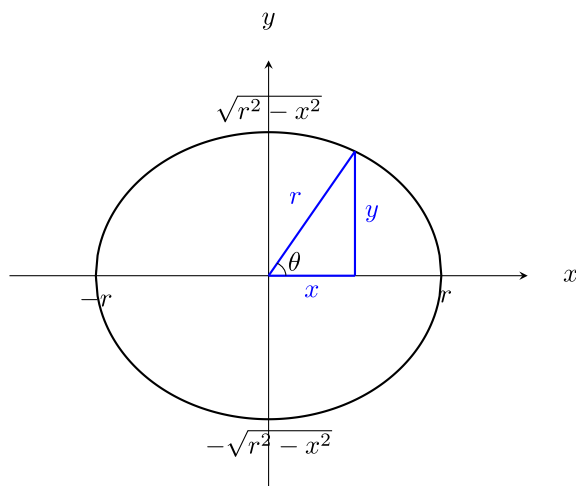
7.3. Trigonometric Substitutions

7.3.1. Motivation

7.3a Learning Objectives

- I can understand the relationship between right triangles and substitutions.
- I can draw triangles to represent substitutions.
- I can understand when and how to use trigonometric substitutions.

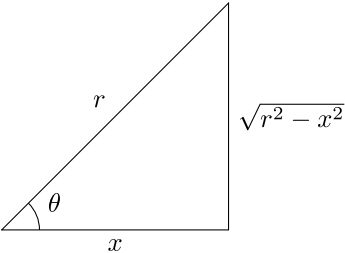
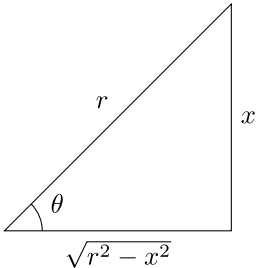
Suppose we want to derive the formula for area of a circle or radius r .



We know how to set up the integral for this using area between curves:

$$\begin{aligned} A &= \int_{-r}^r \underbrace{\sqrt{r^2 - x^2}}_{\text{top}} - \underbrace{(-\sqrt{r^2 - x^2})}_{\text{bottom}} dx \\ &= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \end{aligned}$$

Up until now, we don't have a method to evaluate this integral. Looking at a point on the circle we can use trig functions to rewrite the expressions in x and y into r and θ . We have a few options, so let's do both and compare.

Step	Cosine	Sine
substitution	$x = r \cos(\theta)$	$x = r \sin(\theta)$
triangle		
differential	$dx = -r \sin(\theta) d\theta$	$dx = r \cos(\theta) d\theta$
bounds for θ	$x = -r \Rightarrow \theta = \pi$ $x = r \Rightarrow \theta = 0$	$x = -r \Rightarrow \theta = 0$ $x = r \Rightarrow \theta = \frac{\pi}{2}$
integral after trig sub steps below!	$2 \int_{\pi}^0 r \sin(\theta)(-r \sin(\theta)) d\theta$	$2 \int_0^{\frac{\pi}{2}} r \cos(\theta)r \cos(\theta) d\theta$
integral after trig identity	$2r^2(-1) \int_{\pi}^0 \frac{1}{2}(1 - \cos(2\theta)) d\theta$	$2r^2 \int_0^{\frac{\pi}{2}} \frac{1}{2}(1 + \cos(2\theta)) d\theta$
Observations	• bounds are flipped • extra negatives	• the bounds are least to greatest • there are no extra negatives

For $x = r \sin(\theta)$ we get $dx = r \cos(\theta) d\theta$. We have to solve little equations for θ to get our new bounds.

$$-r = r \sin(\theta) \Rightarrow \theta = \pi$$

$$r = r \sin(\theta) \Rightarrow \theta = 0$$

Then the integral becomes

$$\begin{aligned}
A &= 2 \int_{-r}^r \sqrt{r^2 - x^2} \, dx \\
&= 2 \int_0^\pi \sqrt{r^2 - (r \sin(\theta))^2} (r \cos(\theta)) \, d\theta \\
&= 2 \int_0^\pi \sqrt{r^2 - r^2 \sin^2(\theta)} (r \cos(\theta)) \, d\theta \\
&= 2 \int_0^\pi \sqrt{r^2(1 - \sin^2(\theta))} (r \cos(\theta)) \, d\theta \\
&= 2 \int_0^\pi \sqrt{r^2 \cos^2(\theta)} (r \cos(\theta)) \, d\theta \\
&= 2 \int_0^\pi r |\cos(\theta)| (r \cos(\theta)) \, d\theta \\
&= 2 \int_0^\pi r^2 \cos^2(\theta) \, d\theta \\
&= 2 \int_0^\pi r^2 \left[\frac{1}{2}(1 + \cos(2\theta)) \right] \, d\theta \\
&= r^2 \int_0^\pi (1 + \cos(2\theta)) \, d\theta
\end{aligned}$$

Then we can finish this with a quick u -substitution, where $u = 2\theta$ and $du = 2 \, d\theta$ and bounds become $u(0) = 0$ and $u(\pi) = 2\pi$. Then

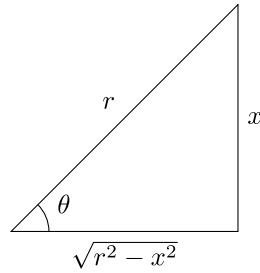
$$\begin{aligned}
A &= \frac{1}{2} r^2 \int_0^{2\pi} (1 + \cos(u)) \, du \\
&= \frac{1}{2} r^2 (u + \sin(u)) \Big|_0^{2\pi} \\
&= \frac{1}{2} r^2 (2\pi + 0 - 0 + 0) \\
&= \pi r^2
\end{aligned}$$

And so we can get the area of the circle! You can try doing the other substitution if you want, using the trigonometric identity $\frac{1}{2}(1 - \cos(2\theta)) = \sin^2(\theta)$.

Ultimately, either substitution **could** work **if you are careful enough**. However, we will just show the substitutions and patterns for the trig subs that don't give extra minus signs. If you are still confused on why we do things in the order we do them, please come chat in office hours about it! (It honestly took me a while to figure this out myself!)

Here is a "more elegant" way to solve this problem, by which we mean a way that takes a lot fewer steps and uses the triangle a bit more:

We set up the integral the same way, but using trigonometric relations and the triangle reproduced here



We can write

$$\cos(\theta) = \frac{\sqrt{r^2 - x^2}}{r} \Rightarrow \sqrt{r^2 - x^2} = r \cos(\theta)$$

Combining this with the differential, $dx = r \cos(\theta) d\theta$, from before, we can jump straight to

$$\begin{aligned} A &= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \\ &= 2 \int_0^\pi r \cos(\theta) r \cos(\theta) d\theta \\ &= 2r^2 \int_0^\pi \cos^2(\theta) d\theta \end{aligned}$$

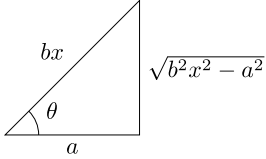
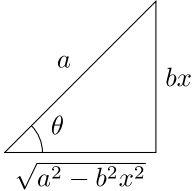
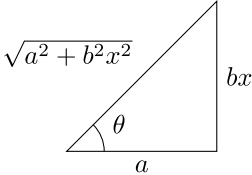
And from this point we finish the problem similarly to before.

Note

This was a lot of writing, but I hope you found a method in here that makes sense, and you understand that there are multiple ways to do the following problems. We will try to just show you the streamlined version of this in future examples, but feel free to try other substitutions if you want the practice!

7.3.2. Substitutions To Use

The substitutions we will use (to avoid extra negatives) for this section will be

Form	Trig Identity	Substitution	Assumptions	Triangle
$\sqrt{b^2x^2 - a^2}$	$\sec^2(\theta) - 1 = \tan^2(\theta)$	$x = \frac{a}{b} \sec(\theta)$	$0 \leq \theta < \frac{\pi}{2},$ $\frac{\pi}{2} < \theta \leq \pi$	
$\sqrt{a^2 - b^2x^2}$	$1 - \sin^2(\theta) = \cos^2(\theta)$	$x = \frac{a}{b} \sin(\theta)$	$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$	
$\sqrt{a^2 + b^2x^2}$	$\tan^2(\theta) + 1 = \sec^2(\theta)$	$x = \frac{a}{b} \tan(\theta)$	$-\frac{\pi}{2} < \theta < \frac{\pi}{2}$	

Note: Most of the examples you will see in this class will have $b = 1$, but the above formulas are given for completion.

7.3.3. Indefinite Integrals

Let's do some examples.

Example 7.3.1

Evaluate

$$\int \frac{x^3}{\sqrt{4 + x^2}} dx$$

Solution

We see a weird root in the integral, so we know to try a trig sub. In this case, the form is $\sqrt{4 + x^2} \rightsquigarrow \sqrt{a^2 + b^2x^2}$ where $a = 2$ and $b = 1$. Then, in this form the substitution we use is

$$x = \frac{a}{b} \tan(\theta)$$

$$x = 2 \tan(\theta)$$

$$dx = 2 \sec^2(\theta) d\theta$$

This substitution gives us

$$\begin{aligned}
\int \frac{x^3}{\sqrt{4+x^2}} dx &= \int \frac{(2 \tan(\theta))^3}{\sqrt{4 + (2 \tan(\theta))^2}} 2 \sec^2(\theta) d\theta \\
&= 16 \int \frac{\tan^3(\theta)}{\sqrt{4(1 + \tan^2(\theta))}} \sec^2(\theta) d\theta \\
&= \frac{16}{2} \int \frac{\tan^3(\theta)}{\sqrt{\sec^2(\theta)}} \sec^2(\theta) d\theta \\
&= 8 \int \tan^3(\theta) \sec(\theta) d\theta, \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}
\end{aligned}$$

Now we have converted the integral into a trigonometric integral that we can solve using methods from Section 7.2. Now, we keep one factor of $\tan(\theta) \sec(\theta)$ and rewrite everything else in terms of $\sec(\theta)$.

$$\begin{aligned}
\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int \tan^2(\theta) \tan(\theta) \sec(\theta) d\theta \\
&= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta
\end{aligned}$$

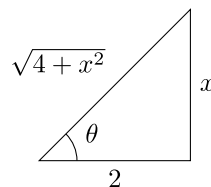
Now we have converted the trig integral into one we can solve with a u -sub where $u = \sec(\theta)$ so $du = \sec(\theta) \tan(\theta) d\theta$. Thus,

$$\begin{aligned}
\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta \\
&= 8 \int u^2 - 1 du \\
&= 8 \left[\frac{u^3}{3} - u \right] \\
&= 8 \left[\frac{\sec^3(\theta)}{3} - \sec(\theta) \right]
\end{aligned}$$

Lastly, the question was in x , not θ . So we need something to replace $\sec(\theta)$ with. Looking back at our substitution, we have

$$x = 2 \tan(\theta) \Rightarrow \tan(\theta) = \frac{x}{2} = \frac{\text{opp}}{\text{adj}}$$

lets us build the triangle below.



So we can see that

$$\sec(\theta) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{4+x^2}}{2}.$$

Finally the antiderivative is

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \left[\frac{\sec^3(\theta)}{3} - \sec(\theta) \right] \\ &= 8 \left[\frac{1}{3} \left(\frac{\sqrt{4+x^2}}{2} \right)^3 - \frac{\sqrt{4+x^2}}{2} \right] + C\end{aligned}$$

Note

That was a really long problem. We can shorten some of the substitutions using the triangle at the end a bit sooner. Here is another way to solve the same problem:

Solution

We see a weird root in the integral, so we know to try a trig sub. In this case, the form is $\sqrt{4+x^2} \rightsquigarrow \sqrt{a^2+b^2x^2}$ where $a=2$ and $b=1$. Then, in this form the substitution we use is

$$x = \frac{a}{b} \tan(\theta)$$

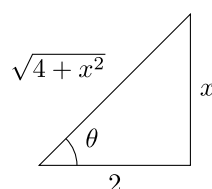
$$x = 2 \tan(\theta)$$

$$dx = 2 \sec^2(\theta) d\theta$$

Further, we have

$$x = 2 \tan(\theta) \Rightarrow \tan(\theta) = \frac{x}{2} = \frac{\text{opp}}{\text{adj}}$$

lets us build the triangle below



Then, we can see that the square root term in our integral can be quickly replaced by a trig function:

$$\sec(\theta) = \frac{\sqrt{4+x^2}}{2} \Rightarrow \sqrt{4+x^2} = 2 \sec(\theta)$$

This substitution gives us

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= \int \frac{(2 \tan(\theta))^3}{2 \sec(\theta)} 2 \sec^2(\theta) d\theta \\ &= 8 \int \tan^3(\theta) \sec(\theta) d\theta, \text{ for } -\frac{\pi}{2} < \theta < \frac{\pi}{2}\end{aligned}$$

Now we have converted the integral into a trigonometric integral that we can solve using methods from Section 7.2. Now, we keep one factor of $\tan(\theta) \sec(\theta)$ and rewrite everything else in terms of $\sec(\theta)$.

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int \tan^2(\theta) \tan(\theta) \sec(\theta) d\theta \\ &= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta\end{aligned}$$

Now we have converted the trig integral into one we can solve with a u -sub where $u = \sec(\theta)$ so $du = \sec(\theta) \tan(\theta) d\theta$. Thus,

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \int (\sec^2(\theta) - 1) \tan(\theta) \sec(\theta) d\theta \\ &= 8 \int u^2 - 1 du \\ &= 8 \left[\frac{u^3}{3} - u \right] \\ &= 8 \left[\frac{\sec^3(\theta)}{3} - \sec(\theta) \right]\end{aligned}$$

Lastly, the question was in x , not θ . So we need something to replace $\sec(\theta)$ with. Looking back at our triangle, we can see that

$$\sec(\theta) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{4+x^2}}{2}.$$

Finally the antiderivative is

$$\begin{aligned}\int \frac{x^3}{\sqrt{4+x^2}} dx &= 8 \left[\frac{\sec^3(\theta)}{3} - \sec(\theta) \right] \\ &= 8 \left[\frac{1}{3} \left(\frac{\sqrt{4+x^2}}{2} \right)^3 - \frac{\sqrt{4+x^2}}{2} \right] + C\end{aligned}$$

Note

A lot of the steps are the same, use whichever method above you prefer when solving these, but always draw the triangle as part of your work.

Example 7.3.2

Evaluate

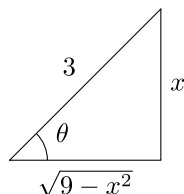
$$\int \frac{\sqrt{9-x^2}}{x^2} dx$$

Solution

The form of the root in this problem is $\sqrt{9-x^2} \rightsquigarrow \sqrt{a^2-b^2x^2}$. So, we know that we need to use the substitution

$$\begin{aligned}x &= \frac{a}{b} \sin(\theta) \\&= 3 \sin(\theta) \\dx &= 3 \cos(\theta) d\theta\end{aligned}$$

The triangle is then



So, the square root part of the integral becomes $\sqrt{9-x^2} = 3 \cos(\theta)$. The substitution gives us

$$\begin{aligned}\int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{3 \cos(\theta)}{(3 \sin(\theta))^2} 3 \cos(\theta) d\theta \\&= \int \cot^2(\theta) d\theta\end{aligned}$$

Now we can try solving this.

$$\begin{aligned}\int \cot^2(\theta) d\theta &= \int \csc^2(\theta) - 1 d\theta \\&= \int \csc^2(\theta) d\theta - \int 1 d\theta \\&= -\cot(\theta) - \theta\end{aligned}$$

Lastly, we need to get back to x for the final answer. Using the triangle from before we can see

$$\cot(\theta) = \frac{\text{adj}}{\text{opp}} = \frac{\sqrt{9-x^2}}{x}$$

and (using any trig function, we just picked sine because we like sine/cosine and it gives a simple expression without a root)

$$\sin(\theta) = \frac{x}{3} \Rightarrow \theta = \arcsin\left(\frac{x}{3}\right)$$

Thus, our final answer is

$$\int \frac{\sqrt{9-x^2}}{x^2} dx = -\frac{\sqrt{9-x^2}}{x} - \arcsin\left(\frac{x}{3}\right).$$

7.3a Section Summary:

- We figured out which trigonometric substitutions would help us simplify problems without extra negatives.
- We used trig substitutions to simplify indefinite integrals.
- We used right triangles to rewrite answers in terms of the original variable.

7.3b Learning Objectives

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7.3.4. Definite Integrals

We touched on it in our motivating example, but we have to be more careful with bounds on these substitutions. Of course, you can find the indefinite integral and deal with bounds separately if you are careful with notation. Here we will go through an example or 2 of a definite integral where we change the bounds. This saves a lot of writing, and means we don't have to rewrite the integral in terms of x at the end.

Example 7.3.3

Find

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx$$

Solution

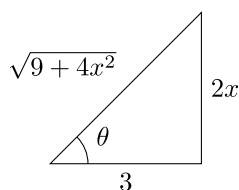
First we rewrite the integrand with something that has a root.

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(4x^2 + 9)^{\frac{3}{2}}} dx = \int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(\sqrt{4x^2 + 9})^3} dx$$

Now we draw the triangle. Since the root has a plus sign (instead of minus) the root term goes on the hypotenuse of the triangle. That means the $2x$ is either on the opposite or the adjacent side, and the trig function that avoids the root is tangent. So the substitution we need to use is

$$x = \frac{3}{2} \tan(\theta)$$
$$\frac{2x}{3} = \tan(\theta) = \frac{\text{opp}}{\text{adj}}$$

. Now we can tell the opposite side is $2x$ and the adjacent side is 3. The triangle is



Ok, now to get the differential

$$x = \frac{3}{2} \tan(\theta)$$
$$dx = \frac{3}{2} \sec^2(\theta) d\theta$$

Lastly we need to rewrite the root expression as a trig function. The trigonometric function that includes the root term and the constant term in our triangle is cosine or secant. We know we already have secant in our problem, so we'll use that

$$\sec(\theta) = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{9+4x^2}}{3}$$

$$\Rightarrow 3\sec(\theta) = \sqrt{9+4x^2}$$

Now we have to actually make this substitution. Plugging in x , dx , and $\sqrt{9+4x^2} = 3\sec(\theta)$ gives us

$$\int_0^{\frac{3\sqrt{3}}{2}} \frac{x^3}{(\sqrt{4x^2+9})^3} dx = \int_{\square}^{\square} \frac{\left(\frac{3}{2}\tan(\theta)\right)^3}{(3\sec(\theta))^3} \left(\frac{3}{2}\sec^2(\theta)\right) d\theta.$$

And the new bounds are found from

$$0 = \frac{3}{2}\tan(\theta) \Rightarrow \theta = 0$$

$$\frac{3\sqrt{3}}{2} = \frac{3}{2}\tan(\theta) \Rightarrow \sqrt{3} = \tan(\theta) \Rightarrow \theta = \frac{\pi}{3}$$

So now we need to solve the following integral

$$\begin{aligned} \int_0^{\frac{\pi}{3}} \frac{\left(\frac{3}{2}\tan(\theta)\right)^3}{(3\sec(\theta))^3} \left(\frac{3}{2}\sec^2(\theta)\right) d\theta &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\tan^3(\theta)}{\sec(\theta)} d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\sin^3(\theta)\cos(\theta)}{\cos^3(\theta)} d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \tan^2(\theta)\sin(\theta) d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} (\sec^2(\theta) - 1)\sin(\theta) d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \frac{\sin(\theta)}{\cos(\theta)} \sec(\theta) - \sin(\theta) d\theta \\ &= \frac{3}{16} \int_0^{\frac{\pi}{3}} \tan(\theta)\sec(\theta) d\theta - \frac{3}{16} \int_0^{\frac{\pi}{3}} \sin(\theta) d\theta \\ &= \frac{3}{16} [\sec(\theta)]_0^{\frac{\pi}{3}} - \frac{3}{16} [-\cos(\theta)]_0^{\frac{\pi}{3}} \\ &= \frac{3}{16} [2 - 1] + \frac{3}{16} \left[\frac{1}{2} - 1\right] \\ &= \frac{3}{32} \end{aligned}$$

Example 7.3.4

Evaluate the following definite integral. Include a drawing of the triangle used in the trig sub.

$$\int_0^1 \sqrt{x - x^2} \, dx$$

Key Solution

We can use trig subs to help us solve even more complicated integrals! We've seen we need something with x squared and something without x .

Recall how to complete the square.

$$\begin{aligned} x - x^2 &= b - (x - c)^2 \\ &= b - (x^2 - 2xc + c^2) \\ &= (b - c^2) + 2cx - x^2 \end{aligned}$$

So

$$2c = 1 \quad \text{and} \quad b - c^2 = 0$$

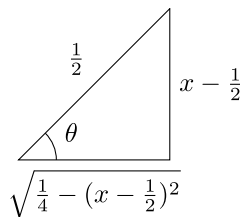
then

$$c = \frac{1}{2} \quad \text{and} \quad b = \frac{1}{4}$$

So we can rewrite

$$\int_0^1 \sqrt{x - x^2} \, dx = \int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} \, dx$$

Setting up the triangle, the hypotenuse is $\frac{1}{4}$. We know the adjacent side to be the square root term, and the opposite side to be $x - \frac{1}{2}$. Then the triangle is



The substitution we use is $\frac{1}{2} \sin(\theta) + \frac{1}{2} = x$ so $dx = \frac{1}{2} \cos(\theta) \, d\theta$, and $\sec(\theta) = \frac{\sqrt{\frac{1}{4} - (x - \frac{1}{2})^2}}{\frac{1}{2}}$. The integrand becomes

$$\int_0^1 \sqrt{\frac{1}{4} - \left(x - \frac{1}{2}\right)^2} \, dx = \int_{\square}^{\square} \frac{1}{2} \sec(\theta) \frac{1}{2} \cos(\theta) \, d\theta$$

The bounds become

$$0 = \frac{1}{2} \sin(\theta) + \frac{1}{2}$$

$$\sin(\theta) = -1$$

$$\theta = -\frac{\pi}{2}$$

and

$$1 = \frac{1}{2} \sin(\theta) + \frac{1}{2}$$

$$\sin(\theta) = 1$$

$$\theta = \frac{\pi}{2}$$

So the integral is an easy one it is

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4} d\theta = \frac{\pi}{4}$$

Tip

Procedure for Trigonometric Substitution

1. Draw a right triangle with appropriate side lengths and angle θ labeled. Write down $\sin(\theta)$, $\cos(\theta)$, and $\tan(\theta)$, in terms of the side lengths.
2. Write down the substitution for the variable x , calculate the differential dx , and specify the selected values of θ for substitution.
3. Substitute the trigonometric expression and the calculated differential into the integrand. Simplify the results algebraically.
4. Integrate the trigonometric integral, keeping in mind the restrictions on the angle θ for reversibility.
5. Use the right triangle to reverse the substitution in the integration result and convert it back to the original variable x . In particular, compositions of trig and inverse trig functions of x should be avoided (final answers should be in fully simplified form).

7.3b Section Summary:

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